Momentum Across Market Regimes – A Machine Learning Approach

By

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A thesis submitted in partial fulfillment of the requirements for the degree of Master of Science in

Quantitative Finance and Actuarial Science

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Date: June 2026

Acknowledgements

I would like to thank my supervisor, Denis Kojevnikov, for his guidance and feedback throughout this project. I am also grateful to Tilburg University's School of Economics and Management for providing access to WRDS data. Finally, I thank my family and friends for their support throughout my studies. Any remaining errors are my own.

AI Disclosure. In accordance with the MSc BAOR/EME/QFAS *Policy on the use of AI Tools for written assignments* (November 2024), the author discloses the following uses of Claude (Anthropic) during the preparation of this thesis.

- **Proofreading and clarity.** Grammar correction, copy-editing, and suggestions for improving the readability of the author's drafts.
- **Idea generation.** Brainstorming structural framings, exploring alternative phrasings, and surfacing candidate literature.
- **Code support.** Code review, refactoring suggestions, and debugging assistance for the analytical scripts.
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Abstract

Cross-sectional momentum, the tendency for recent stock-market winners to keep winning and losers to keep losing, is one of the most profitable anomalies in finance and one of the most fragile. Momentum portfolios suffered losses exceeding 50% during the 2009 market rebound and have crashed repeatedly around regime transitions. The standard responses scale exposure during stress or blend horizon weights, leaving the selection rule itself unconditional. This thesis asks a different question: how does the cross-sectional structure of momentum reorganize between calm and panic regimes, and what mechanism drives that reorganization?

The empirical framework has two stages. A Bayesian two-state Hidden Markov Model, estimated by Gibbs sampling on four monthly macrofinancial stress indicators, produces a continuous filtered probability of market panic that updates each month. This regime signal is then combined with twelve momentum horizons inside a cross-sectional stock-selection model (an XGBoost gradient-boosted regressor, benchmarked against a deterministic formula and a logistic regression). Stocks are ranked monthly and the top and bottom deciles enter a value-weighted long-short portfolio. All models are estimated on 1990–2010 U.S. equities and evaluated out-of-sample on the 167-month 2011–2024 window, with international replication on UK and Japanese equities. The combination of an HMM regime signal with an XGBoost cross-sectional model is, to the author's knowledge, novel, and extracts a stronger momentum signal than conventional regime indicators such as market drawdown.

This thesis develops a more flexible regime-dependent momentum framework that optimizes portfolio weights directly from stock-level signals. The portfolio selected is not restricted to forming portfolios from pre-specified momentum rankings, whether by selecting the top-ranked stocks within a single horizon, rotating across a small set of predefined horizons, or lowering exposure when market conditions deteriorate. Instead, the model learns how stock-level signals, and not horizon level signals, should translate into portfolio positions in each regime. The empirical results show that the learned rule behaves conventionally in calm regimes, allocating the long leg to recent winners. However, in panic regimes, the long leg is not simply drawn from the winners of another horizon. It is rebuilt from a distinct set of stocks not top-ranked at any individual momentum horizon, a novel form of selection whose portfolio relevance emerges only after conditioning on the panic state and combining information across signals.

Contents

1 Introduction

This thesis applies machine learning to study the behavior of cross-sectional momentum[1] across market regimes. The aim is to better understand this anomaly conditional on the market regime, not to propose a strategy that outperforms the market on aggregate.

Cross-sectional momentum is one of the most robust anomalies in empirical finance and one of the most fragile. This fragility motivates a substantial literature on momentum-crash management, but existing responses share a structural assumption. The regime signal enters only to scale or re-weight a fixed scoring rule, never as a feature inside the rule itself. Barroso and Santa-Clara (2015) target constant volatility during stress, while Cooper et al. (2004) treat regimes as a binary up-or-down variable and Goulding et al. (2023) blend fast and slow horizon weights. This leaves unexamined how the full momentum term structure and the regime signal jointly determine which stocks the strategy selects in each market state.

This thesis examines that possibility. The regime signal acts as a *context variable*. It carries no direct return-predictive content, but conditions how a nonlinear cross-sectional model uses the momentum term structure in a given market state.

Research question

The thesis asks how the cross-sectional structure of momentum changes between calm and panic regimes, and what selection rule drives that change.

Approach

The question is answered in two steps. First, a Bayesian Hidden Markov Model (HMM) is estimated on four monthly macrofinancial stress indicators (drawdown, return dispersion, relative market participation, and credit spread), producing a filtered probability of market panic that updates monthly. This regime signal is combined with the twelve momentum horizons inside an XGBoost cross-sectional stock-selection model, benchmarked against a deterministic formula and a logistic regression, and evaluated out-of-sample on 2011–2024 U.S. equities in long-short construction. Second, the XGBoost selection rule is decomposed month by month with a per-feature contribution analysis and a clustering of long-leg cross-sectional z-curves, identifying the role the regime signal plays in selection.

Headline findings

On 2011–2024 U.S. equities, the XGBoost ensemble on the regime probability and the twelve momentum horizons produces a six-factor alpha of 24.1% ($t = 4.81$). A linear baseline on identical features collapses, and removing the regime probability from XGBoost alone cuts its Sharpe by more than 60%. The mechanism is that the cross-section

[1] Cross-sectional momentum ranks stocks against each other at each point in time, going long the relative winners and short-selling the relative losers within the same universe (Jegadeesh and Titman, 1993).

of momentum reorganizes systematically across regimes. In calm the model selects past winners in the classical sense; in panic the regime signal flips the selection direction, with the long leg drawn from the most beaten-down names while the long-short portfolio stays fully invested. The long leg holds stocks not top-ranked at any individual momentum horizon, and the long-leg market beta stays above the short-leg beta in every regime, in contrast to the crash mechanism documented for unconditional momentum. Cluster analysis recovers four recurring selection modes that trace a single market cycle.

Contributions

The thesis makes two contributions to the momentum literature. First, the methodological combination of a continuous probabilistic regime filter (the Bayesian HMM filtered probability of market panic) with a nonlinear cross-sectional model (XGBoost) is novel relative to the linear and rank-based regime-conditional rules in the prior literature, and it enables the empirical characterization developed below. Second, the empirical analysis characterizes how the regime-momentum mechanism manifests at a granularity the prior literature does not provide, decomposing XGB's monthly picks into four recurring selection modes and showing that the regime signal acts as an irreducibly nonlinear routing variable for the momentum term structure. The calm-panic asymmetry itself was noted descriptively by Cooper et al. (2004); this thesis implements the regime-conditioned response concretely as a single nonlinear model routing on a continuous real-time regime probability, with the long-short portfolio fully invested throughout.

Outline

Section 2 (Literature Review) covers momentum's horizon structure, momentum crashes and exposure-based risk management, regime models in asset pricing, and machine learning in cross-sectional finance. Section 3 (Methodology) develops the two-stage framework, including the Bayesian HMM specification and Gibbs sampling estimation, and the cross-sectional stock-selection methods. Section 4 (Data) describes the U.S. equity sample, the four-pass HMM feature selection pipeline, and the cross-sectional features. Section 5 (Results) presents the out-of-sample results, the cross-sectional z-score and cluster analysis that documents the regime-momentum mechanism, robustness and risk-aversion analyses, and international replication. Section 6 (Conclusion) summarizes the contributions, discusses limitations, and suggests directions for future research.

2 Literature Review

2.1 The Momentum Anomaly and Its Horizon Structure

Jegadeesh and Titman (1993) established cross-sectional momentum as a pervasive feature of equity markets. Portfolios that buy past winners and sell past losers earn significant returns over 3-to-12-month holding periods. They examine sixteen formation-and-holding-period combinations from three to twelve months, with the twelve-month formation period emerging as the most profitable. The strategy has since been replicated across international markets, asset classes, and sample periods (Jegadeesh and Titman, 2001; Asness et al., 2013).

The economic mechanism behind momentum remains debated. Behavioral explanations include gradual information diffusion (Hong and Stein, 1999) in addition to the joint conservatism representativeness mechanism of Barberis et al. (1998), while risk-based explanations link momentum to time-varying systematic risk. This thesis does not take a position on the underlying mechanism but examines the empirical regularity that momentum's cross-sectional structure varies with market conditions.

A defining feature of momentum is its sensitivity to the lookback horizon. Short lookbacks of one to three months capture short-term reversal effects (Jegadeesh, 1990; Lehmann, 1990), intermediate horizons (months 7 to 12 prior to the formation month) capture continuation (Novy-Marx, 2012), and lookbacks beyond twelve months reflect long-term mean reversion (De Bondt and Thaler, 1985). Lee and Swaminathan (2000) document a momentum lifecycle in which the same signal evolves from continuation to reversal as information ages. Together these findings suggest that momentum is a family of horizon-specific patterns whose relative strength varies with market conditions, not a single signal. The framework here tests this directly by exposing all twelve monthly horizons jointly to the cross-sectional model rather than committing to a single lookback ex ante.

2.2 Momentum Crashes and Exposure-Based Solutions

Despite its long-run profitability, momentum is prone to severe crashes. Daniel and Moskowitz (2016) document that momentum portfolios suffered catastrophic losses during the 1932 and 2009 market rebounds, and show that these crashes are predictable. When the market has recently declined, past losers tend to be high-beta stocks that rally disproportionately during recoveries, while past winners are low-beta defensive stocks that lag. In economic terms, momentum is a bet that the recent regime persists. It earns the continuation premium when the regime holds, and crashes at regime transitions, when the cross-sectional ordering inherited from the prior regime becomes the wrong bet for the new one. Barroso and Santa-Clara (2015) propose a simple risk-management response, scaling momentum exposure inversely with the strategy's recent realized volatility, a constant-volatility overlay that substantially raises the Sharpe ratio. Bianchi et al. (2022) pursue a related approach by constructing a crash indicator from the joint time variation of conditional volatility and skewness, again operating on the strategy's exposure rather than its composition. This thesis explores a complementary composition-side route: rather than scaling exposure during stress, the model optimizes on which stocks populate each leg conditional on the regime. As a consequence, the cross-section is re-ranked across regimes and the long-leg market beta remains above the short-leg beta in every regime (Appendix H.5).

The liquidity literature provides a theoretical basis for why momentum crashes reverse. Margin spirals (Brunnermeier and Pedersen, 2009) and institutional flow pressure (Vayanos and Woolley, 2013) create temporary price dislocations during stress that revert when selling pressure subsides, the mechanism central to the regime-momentum interpretation developed below.

2.3 Regime Models and Regime-Conditional Asset Pricing

Hidden Markov Models have a long history in econometrics. The probabilistic machinery dates to Baum et al. (1970); Hamilton (1989) introduced the regime-switching tradition with a business-cycle application to U.S. GNP growth, and Bayesian estimation via Gibbs sampling was developed by Albert and Chib (1993) with the multivariate-emission generalization now standard (Frühwirth-Schnatter, 2006). In the spirit of Hamilton's approach of fitting the regime model to macroeconomic observables rather than to returns directly, this thesis fits the regime model to macrofinancial stress indicators.

A separate literature asks whether asset-pricing anomalies are themselves regime-conditional. Cooper et al. (2004) establish that momentum profits depend on the prior market state, with strong continuation following up-markets and weaker or reversed performance following down-markets. More recent work integrates regime estimation into portfolio construction, for example Shu and Mulvey (2024), who feed sparse-jump-model factor regimes into a Black-Litterman allocation.

The most directly related work is Goulding et al. (2023), who classify each month into one of four market cycles (Bull, Correction, Bear, Rebound) from the trailing one-month and rolling twelve-month market returns and blend the two momentum horizons via a cycle-specific weight $\alpha \cdot mom_1 + (1 - \alpha) \cdot mom_{12}$. The framework here generalizes that approach along three axes: from a deterministic four-state classification based on trailing returns to a continuous panic probability from a Bayesian HMM on real-time stress indicators; from two horizons to the full twelve-horizon term structure; and from a parametric linear blend to a learned nonlinear combination. Their strategies are a time-series optimization of a single market index rather than a cross-sectional long-short of individual stocks, so a direct performance comparison is not appropriate.

The unifying feature of this subsection is that the regime variable enters portfolio decisions through a hand-coded rule, threshold, or parametric weight, never as a continuous learned feature. This thesis instead supplies the probability of being in a calm/panic regime as a feature to a nonlinear cross-sectional model, which learns how to use it to optimize portfolio construction.

2.4 Machine Learning in Cross-Sectional Asset Pricing

The use of machine learning methods for cross-sectional stock selection has grown rapidly. Gu et al. (2020) compare a wide range of methods on a kitchen-sink characteristic set including multiple momentum horizons, finding that tree-based models and neural networks substantially outperform linear models by capturing nonlinear interactions among firm characteristics. Gradient-boosted trees (Chen and Guestrin, 2016) have proven particularly effective for tabular financial data because they handle missing values natively, capture nonlinear relationships and interactions without explicit specification, and are robust to irrelevant features. A central challenge in applying these methods to asset pricing is interpretability. This thesis addresses it on two fronts: SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020) attribute each prediction to feature-level contributions, and a clustering of the long leg's monthly cross-sectional z-curves into recurring selection modes characterizes what the model selects in each regime rather than only which features matter.

The closest methodological precedent is Beckmeyer and Wiedemann (2025), who use machine learning to learn data-driven weights across past daily returns for a single momentum signal, constructing a dynamically reweighted strategy that outperforms the conventional twelve-minus-one specification. The framework here differs in two respects. First, the weighting is learned not over the daily history of one signal but over a vector of twelve monthly horizons. Second, the weighting is learned *conditional on a regime context variable*, in contrast to the unconditional weighting in Beckmeyer and Wiedemann (2025), which averages across calm and panic regimes. Isolating the regime-conditioning role is this framework's primary contribution, and the thesis tests whether disaggregating that average reveals a structural reorganization of the term structure between states.

Despite these advances, to the best of our knowledge no published study examines how the full term structure of cross-sectional momentum predictability reorganizes across regimes, or whether capturing this reorganization requires nonlinear modeling.

3 Methodology

This chapter combines ideas and statistical methods from the literatures reviewed in Section 2 into a two-stage framework that tests the research question posed in Section 1. Stage 1 estimates a Bayesian two-state Hidden Markov Model on a single macrofinancial stress indicator, the market drawdown, producing a continuous probability of the panic state. Stage 2 combines that probability with cross-sectional momentum inside a series of stock-selection models. The case for conditioning on drawdown alone, grounded in the momentum-crash literature and confirmed by a selection rule that uses training information only, is developed in Section 4.1.

3.1 Stage 1: Regime Classification via Hidden Markov Model

3.1.1 Model Specification

This study employs a two-state ($K = 2$) Bayesian Hidden Markov Model to classify the market into latent regimes[2]. The hidden state $s_t \in \{0, 1\}$ represents the unobserved regime at month t, where $s_t = 0$ denotes the *calm* regime and $s_t = 1$ denotes the *panic* regime. The labels *calm* and *panic* reflect the economic nature of the chosen stress feature, which is diagnostic of market stress rather than trend direction, business-cycle phase, or volatility alone. An HMM fitted to a different feature set would identify different regimes. As emphasized by Hamilton (1989), market regimes are not directly observable. An investor cannot know with certainty whether the market is currently in a calm or panic state, and classifications that look obvious in retrospect are genuinely difficult to identify in real time. The HMM formalizes this uncertainty by treating regimes as hidden states that must be inferred probabilistically from observable market data.

Observation equation

Conditional on the regime, the vector of standardized features[3] $\mathbf{z}_t \in \mathbb{R}^D$ is drawn from a regime-specific multivariate normal distribution. We assume that, conditional on the regime $s_t = k$, $\mathbf{z}_t$ is normally distributed with mean vector $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$ for $k \in \{0, 1\}$ (calm and panic). Writing $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for the multivariate normal distribution with the indicated mean and covariance, and using the shorthand $\mathbf{z}_{1:t} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t)$

[2] The choice of $K = 2$ is validated empirically. Appendix G.5 shows that increasing to $K = 3, 4$, or 5 states does not improve downstream portfolio performance and introduces estimation instability.

[3] $\mathbb{R}^D$ denotes the D-dimensional space of real-valued vectors; the D stress indicators at month t are stacked into a single point in this space so the HMM can model them jointly. The framework is stated for general D because the selection exercise of Section 4.1 also estimates multi-feature candidate specifications. The production model uses $D = 1$ (market drawdown only), so the emission distributions reduce to univariate normals.

for the history of observations through month t,

$$(\mathbf{z}_t \mid s_t = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \qquad (\mathbf{z}_t \mid s_t = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1). \tag{1}$$

Each regime is characterized by its own mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^D$ and covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D\times D}$, where $k \in \{0 \text{ (calm)}, 1 \text{ (panic)}\}$. For the multi-feature candidate specifications considered during selection, the multivariate form captures both level differences in individual features across regimes and changes in the correlation structure. In the drawdown-only production model ($D = 1$) it reduces to a regime-specific mean and variance: panic is identified as the state with a deeper average drawdown and more volatile drawdown innovations. The Gaussian assumption is standard in the HMM literature. Appendix H.8 verifies its adequacy by comparing it to heavier-tailed multivariate Student-t emissions.

Transition equation

The evolution of the hidden state follows a Markov chain with a 2×2 transition matrix:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}, \tag{2}$$

where $P_{ij} = P(s_t = j \mid s_{t-1} = i)$ and each row sums to one. The diagonal elements P_{00} (calm persistence) and P_{11} (panic persistence) govern how likely each regime is to continue into the next month. The expected duration of regime k is $1/(1 - P_{kk})$ months, so a persistence of 0.95 corresponds to a regime that lasts roughly 20 months on average.

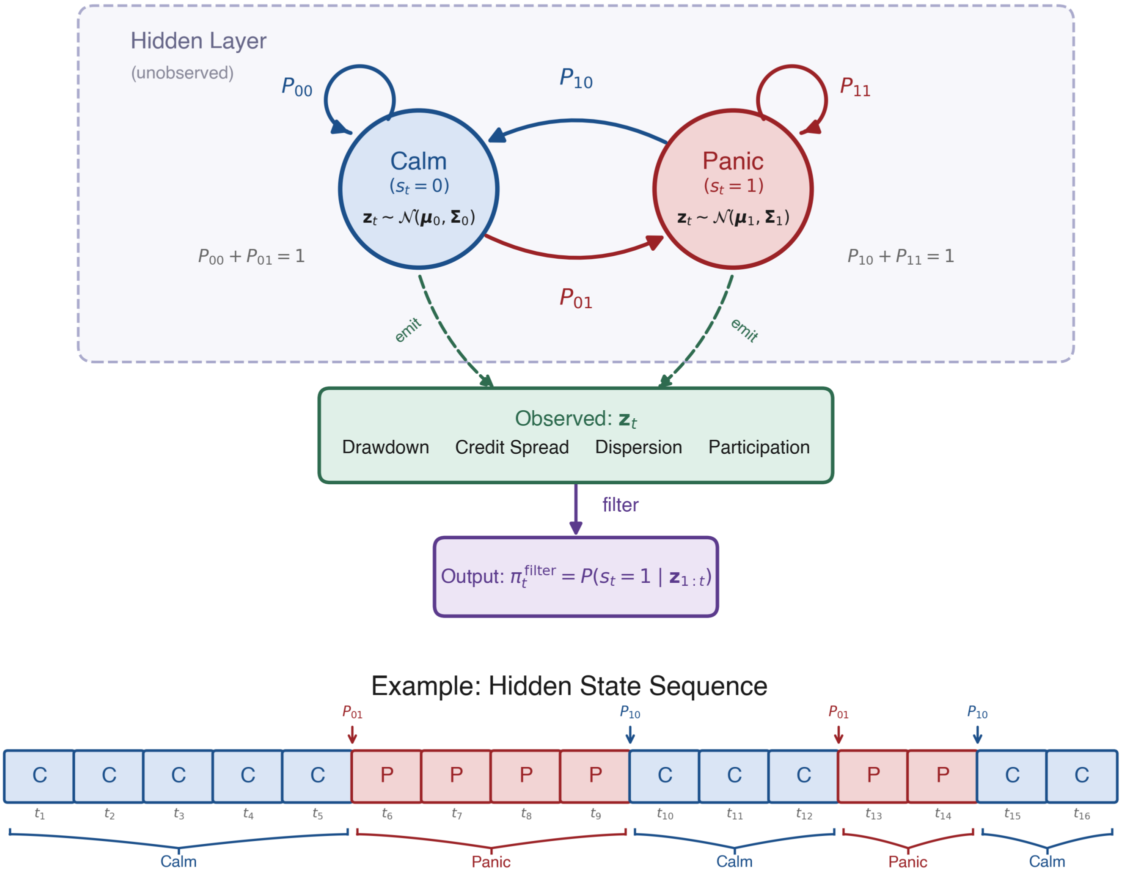

Figure 1: Architecture of the two-state Hidden Markov Model. Hidden states (Calm/Panic) evolve via transition probabilities P_{ij} and emit the observation vector $\mathbf{z}_t \in \mathbb{R}^D$. The forward filter inverts this process to produce $\pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$. The lower panel shows an example state sequence illustrating regime persistence and abrupt transitions.

3.1.2 Bayesian Estimation via Gibbs Sampling

The standard frequentist alternative is the Baum–Welch algorithm (Baum et al., 1970), a special case of Expectation-Maximization (Dempster et al., 1977), which finds maximum likelihood point estimates.

Maximum likelihood estimation of regime-switching HMMs has several known shortcomings for the present application. The likelihood surface is multimodal, with at minimum two equivalent global optima arising from the invariance of the likelihood under state relabeling. Additionally, EM can converge to degenerate solutions in which one regime collapses around a single observation (Frühwirth-Schnatter, 2006). Both problems are exacerbated for rare regimes. The asymptotic standard errors that frequentist inference relies on become unreliable when few observations support the rare-state estimates, and the standard regularity conditions can fail outright: the persistence parameters P_{00} and P_{11} are unidentified when the two regimes' means and covariances coincide, and the information matrix becomes singular at that point (Hamilton, 1989, footnote 12).

The Bayesian approach addresses these issues directly. Posterior sampling integrates over parameter uncertainty rather than relying on a point estimate. Conjugate priors regularize rare-regime estimates with explicit, defensible weights (Section 3.1.3). Finally, convergence diagnostics across multiple chains (Appendix J) diagnose multimodality that would otherwise go unnoticed under a single EM run.

The Bayesian approach instead draws from the joint posterior:

$$p\big(\{s_t\}_{t=1}^{T},\ \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=0}^{1},\ \mathbf{P}\,\big|\,\mathbf{z}_{1:T}\big)\,.$$

The states and the parameters are coupled: we can sample from the distribution of one block given the other, but not from either block on its own. Gibbs sampling (Albert and Chib, 1993; Hoff, 2009), a Markov chain Monte Carlo (MCMC) method, exploits exactly that structure by alternating between three conditional updates over the state path, the emission parameters, and the transition matrix. Conjugate priors ensure that each block can be sampled exactly from its conditional posterior.

3.1.3 Prior Specification

Emission parameters

Before seeing the data, neither regime should be forced to have extreme values in any feature; the model should learn the characteristics of calm and panic states from the data. This is encoded by a conjugate Normal-Inverse-Wishart (NIW) prior on each regime's mean vector and covariance matrix:[4]

$$\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_0, \boldsymbol{\Psi}_0), \qquad \boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}\left(\mathbf{m}_0, \frac{\boldsymbol{\Sigma}_k}{\kappa_0}\right). \tag{3}$$

The hyperparameters are chosen as

$$\mathbf{m}_0 = \mathbf{0}, \qquad \kappa_0 = 0.01, \qquad \nu_0 = D + 2, \qquad \boldsymbol{\Psi}_0 = \mathbf{I}_D(\nu_0 - D - 1), \tag{4}$$

where D is the number of selected features. After observing data, these prior hyperparameters update to posterior counterparts κ_n, $\mathbf{m}_n$, ν_n, and $\boldsymbol{\Psi}_n$ (Equation 9). Each hyperparameter controls a different aspect of the prior belief:

- $\mathbf{m}_0$ is the prior guess for each regime's mean vector. $\mathbf{m}_0 = \mathbf{0}$ is the natural prior

[4] The Inverse-Wishart is a probability distribution over positive-definite matrices, making it the natural prior for a covariance matrix. Every sample is guaranteed to be symmetric and positive-definite. The Normal-Inverse-Wishart combines this with a conditional Normal prior on the mean.

given that the features are standardized. Without data, neither regime is presumed to center away from zero.

- κ_0 controls how strongly the prior mean $\mathbf{m}_0$ is trusted relative to the data. The very small value $\kappa_0 = 0.01$ makes this prior nearly uninformative, so the posterior mean is driven overwhelmingly by the observed data.

- ν_0 sets the Inverse-Wishart prior's degrees of freedom, interpretable as the number of pseudo-observations the prior is worth. The value $\nu_0 = D+2$ is the smallest that yields a proper prior with finite mean; a larger choice tightens the prior around $\mathbf{I}_D$.

- $\mathbf{\Psi}_0$ is the scale matrix of the Inverse-Wishart prior. It is calibrated so that the prior expected covariance equals the identity matrix, $\mathbb{E}[\mathbf{\Sigma}_k] = \mathbf{\Psi}_0/(\nu_0 - D - 1) = \mathbf{I}_D$, implying unit marginal variance and no cross-feature correlation before observing the data.

Transition matrix

Regimes should be persistent rather than switching every few months. This is encoded by assigning each row of the transition matrix an independent Dirichlet prior whose mass concentrates on the diagonal:

$$\mathbf{P}_{0,\cdot} \sim \mathrm{Dir}(9,1), \qquad \mathbf{P}_{1,\cdot} \sim \mathrm{Dir}(1,9). \tag{5}$$

Equivalently, in matrix form,

$$\boldsymbol{\alpha}_{\mathrm{dir}} = \begin{pmatrix} 9 & 1 \\ 1 & 9 \end{pmatrix}.$$

This prior favors persistence in both regimes. Without data, the prior mean transition matrix is

$$\mathbb{E}[\mathbf{P}] = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}. \tag{6}$$

The prior expects each regime to continue with probability 0.9, corresponding to an expected duration of roughly 10 months. It is informative enough to discourage pathological solutions but weak enough to be overridden by the data.

3.1.4 Gibbs Sampling Algorithm

Initialization

The Gibbs sampler requires starting values for the emission parameters and the transition matrix. A crude binary partition based on a single stress indicator is used:[5] months above

[5] The choice of initialization has negligible impact on the posterior. The sampler is run for 2,000 iterations with 500 discarded as burn-in, and convergence diagnostics (Section 4.3, Appendix J) confirm that the chain mixes well regardless of the starting partition.

the indicator's median (the more stressed half) are initially assigned to the panic state ($s_t = 1$), and the rest to the calm state ($s_t = 0$). The initial emission parameters are then computed from each group:

$$\boldsymbol{\mu}_k^{(0)} = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t, \qquad \boldsymbol{\Sigma}_k^{(0)} = \mathrm{Cov}(\{\mathbf{z}_t : s_t = k\}),$$

where n_k is the number of months assigned to regime k. The initial transition matrix is set to the prior mean, $\mathbf{P}^{(0)} = \mathbb{E}[\mathbf{P}]$ (Equation 6). These initial parameters give the first forward pass (described below) emission densities that can distinguish the two states.

Iterative updates

Each iteration cycles through three conditional updates: the latent state path $\{s_t\}_{t=1}^{T}$, the emission parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, and the transition matrix $\mathbf{P}$. Each block is sampled from its full conditional distribution, with the other two blocks held fixed at their current values; for Steps 2 and 3 this distribution takes the conjugate Bayes form posterior $\propto$ likelihood $\times$ prior, while Step 1 (FFBS) samples the state path jointly from its full conditional via the forward-filtering backward-sampling recursion described below. The likelihood factor for Steps 1 and 2 is the Gaussian emission density $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, the plausibility of the observed feature vector $\mathbf{z}_t$ under regime k; Step 3 instead uses the multinomial likelihood of the sampled transition counts n_{ij}. The cycle is summarized graphically below; the steps that follow derive each update in turn.

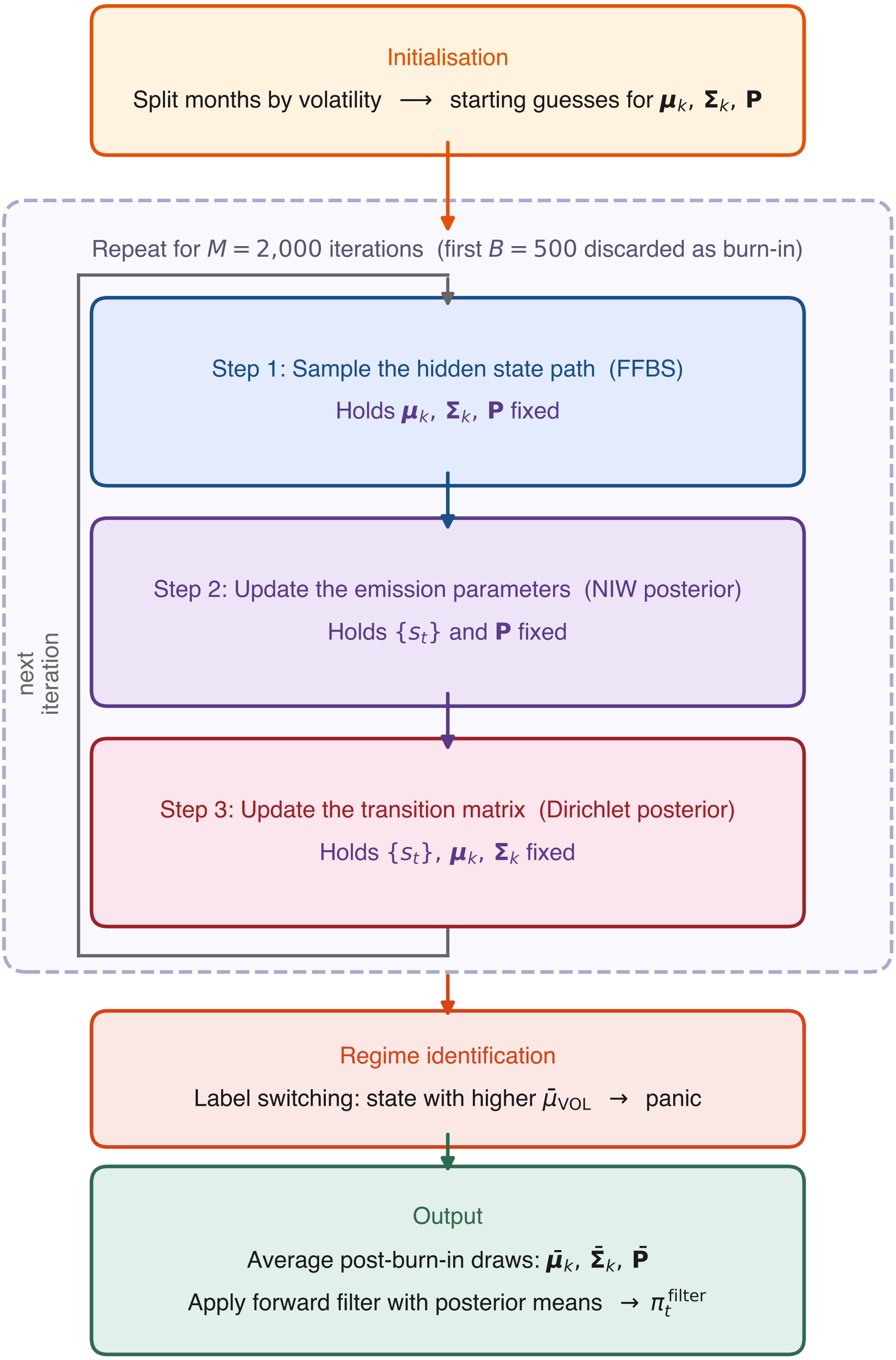

Figure 2: Flowchart of the Gibbs sampler.

Step 1: Sampling the hidden state path via FFBS

The full latent regime sequence $(s_1, \ldots, s_T)$ is sampled using Forward-Filtering Backward-Sampling (FFBS; see Frühwirth-Schnatter, 2006, ch. 11), with the current emission parameters and transition matrix held fixed.

The forward pass (the *forward filter*) computes the filtered probability

$$\alpha_t(k) = P(s_t = k \mid \mathbf{z}_{1:t}),$$

the probability of being in state k at time t given all observations so far. $\alpha_t(k)$ is built in two sub-steps. First, a *prediction step* asks: before seeing this month's data, how likely is each state? Since the previous state is unknown, the algorithm sums over both possibilities, weighting each by its filtered probability from the previous month and the transition probability:

$$\alpha_{t|t-1}(k) = \alpha_{t-1}(0)\, P_{0k} + \alpha_{t-1}(1)\, P_{1k}.$$

Second, an *update step* revises this prediction once the current observation $\mathbf{z}_t$ arrives, using Bayes' rule. If the observed features look more like a panic month than a calm month, the probability of panic increases. Combining the two steps yields the recursion

$$\alpha_t(k) = \frac{f(\mathbf{z}_t \mid s_t = k)\; \alpha_{t|t-1}(k)}{f(\mathbf{z}_t \mid s_t = 0)\; \alpha_{t|t-1}(0) + f(\mathbf{z}_t \mid s_t = 1)\; \alpha_{t|t-1}(1)}, \tag{7}$$

where $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian emission density.[6] The recursion is initialized with a uniform initial state distribution $P(s_1 = k) = 1/2$, reflecting no prior preference for either regime. For the first observation this gives

$$\alpha_1(k) \propto f(\mathbf{z}_1 \mid s_1 = k).$$

The filtered probabilities $\alpha_t(k)$ are marginals. They describe each month's regime in isolation, not how consecutive months connect. Sampling each s_t independently from its α_t would ignore the transition matrix and could produce paths that flip regimes from one month to the next, in conflict with the strongly persistent $\mathbf{P}$. A second, backward pass is therefore needed to draw a full state path that respects $\mathbf{P}$. The backward pass proceeds in two steps:

[6] In practice, these densities can be extremely small (or large) for multi-dimensional observations, leading to numerical underflow. The implementation avoids this by working in log-space. Log-likelihoods $\log f(\mathbf{z}_t \mid s_t = k)$ are computed directly, and the log-sum-exp identity $\log(e^a + e^b) = \max(a, b) + \log(1 + e^{-|a-b|})$ is used whenever probabilities must be normalized.

Terminal state

At $t = T$, the filtered probability already conditions on all observations, so the terminal state is drawn directly:

$$s_T \sim \mathrm{Cat}(\alpha_T(0),\, \alpha_T(1)),$$

that is, the algorithm draws $s_T = 0$ (calm) with probability $\alpha_T(0)$ and $s_T = 1$ (panic) with probability $\alpha_T(1)$.[7]

Backward recursion

For each earlier time step $t = T-1, T-2, \ldots, 1$, the algorithm chooses a state consistent with two pieces of information: (i) the filtered belief $\alpha_t(k)$ from the forward pass, and (ii) the already-sampled future state s_{t+1}, which constrains the current state via the transition probability $P_{k,s_{t+1}}$. Combining these two terms and normalizing gives:

$$P(s_t = k \mid s_{t+1}, \mathbf{z}_{1:t}) = \frac{\alpha_t(k)\, P_{k,s_{t+1}}}{\alpha_t(0)\, P_{0,s_{t+1}} + \alpha_t(1)\, P_{1,s_{t+1}}}. \tag{8}$$

For example, if the forward filter assigns high probability to calm at time t ($\alpha_t(0)$ is large) and the already-sampled s_{t+1} is also calm (so $P_{0,0}$ is large), then the product $\alpha_t(0)\, P_{0,0}$ dwarfs $\alpha_t(1)\, P_{1,0}$, the conditional probability of $s_t = 0$ approaches one, and the algorithm very likely samples $s_t = 0$.

The output of the full FFBS procedure is one draw from the posterior over state paths. Across Gibbs iterations, different paths are sampled, exploring the full posterior uncertainty.

The backward pass is used only during training. The sampled state path it produces feeds the conjugate Gibbs updates of the emission and transition parameters. The trading signal π_t^{filter} relies solely on the forward filter.[8]

Step 2: Updating the emission parameters

Given the sampled state path, each regime k has n_k observations assigned to it. From these, two summary statistics are computed:

$$\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t, \qquad \mathbf{S}_k = \sum_{t:\, s_t = k} (\mathbf{z}_t - \bar{\mathbf{x}}_k)(\mathbf{z}_t - \bar{\mathbf{x}}_k)^\top.$$

[7] Cat denotes the categorical distribution, the standard notation for a single draw from a discrete probability vector. With two states it reduces to a weighted coin flip.

[8] The forward filter uses only observations $\mathbf{z}_{1:t}$ (past and present) at each month t, never future months $\mathbf{z}_{t+1:T}$, so no look-ahead information leaks into the trading signal. The smoothed counterpart $P(s_t \mid \mathbf{z}_{1:T})$, which would condition on the full sample, is not used.

The posterior hyperparameters are:

$$\kappa_n = \kappa_0 + n_k, \qquad \mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_k \bar{\mathbf{x}}_k}{\kappa_n}, \qquad \nu_n = \nu_0 + n_k,$$
$$\boldsymbol{\Psi}_n = \boldsymbol{\Psi}_0 + \mathbf{S}_k + \frac{\kappa_0 n_k}{\kappa_n}(\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^\top. \tag{9}$$

New regime parameters are then drawn from the updated posterior: $\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_n, \boldsymbol{\Psi}_n)$ and $\boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}(\mathbf{m}_n, \boldsymbol{\Sigma}_k/\kappa_n)$.

Step 3: Updating the transition matrix

The sampled path is scanned to count transitions, $n_{ij} = \sum_{t=2}^{T} \mathbb{1}(s_{t-1} = i,\ s_t = j)$, the number of times the path moved from state i to state j. Each row of the transition matrix is then drawn from its own Dirichlet posterior:

$$\mathbf{P}_{i,\cdot} \mid \{s_t\} \sim \mathrm{Dir}\big(\alpha_{i0} + n_{i0},\ \alpha_{i1} + n_{i1}\big). \tag{10}$$

Sampling once from each of the two Dirichlet posteriors (one per regime) produces the updated transition matrix $\mathbf{P}$, which is held fixed during the next iteration's state-path sampling. Because the prior is strongly tilted toward persistence, the sampled $\mathbf{P}$ tracks the data while staying anchored around a persistent diagonal even on short or noisy paths.

With all three blocks updated, the sampler returns to Step 1. After sufficient iterations the draws converge to samples from the joint posterior.

3.1.5 Implementation and Seed Averaging

The Gibbs sampler is re-estimated on the expanding training window each year (Section 4.3). To reduce sensitivity to MCMC initialization, the full sampler is run independently with $N_s = 50$ different random seeds. Each chain is run for $M = 2{,}000$ iterations with the initialization described above. The first $B = 500$ iterations are discarded as burn-in, leaving 1,500 posterior draws per seed.

For each seed, posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ are computed by averaging over the post-burn-in draws. The forward filter is then applied to the full sample using these parameters to obtain a seed-specific filtered probability $\pi_{t,s}^{\text{filter}}$. The final trading signal is the average across seeds:

$$\pi_t^{\text{filter}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \pi_{t,s}^{\text{filter}}. \tag{11}$$

Averaging across MCMC chains is a methodological requirement, not a performance optimization. A single Gibbs chain produces a noisy regime signal whose downstream Sharpe varies meaningfully across random initializations; averaging across chains reduces this noise, and the variance shrinks as more chains are added. By 50 chains the downstream Sharpe has effectively stabilized, with the plateau setting in well below that count (Appendix J.4). All out-of-sample numbers reported below are computed from this 50-chain ensemble, not from a single chain.

3.1.6 Regime Identification and Trading Signal

Label Switching and Regime Naming

In a two-state HMM, the labels 0 and 1 are intrinsically arbitrary. Relabeling the states leaves the likelihood unchanged, so the HMM identifies two clusters in the D-dimensional feature space without knowing which one should be called panic. To break this symmetry, each feature is assigned an expected sign during stress (the direction in which it moves during historical crisis windows relative to calm periods), and the cluster whose posterior mean vector best aligns with that pattern is designated as "panic" and the other as "calm". For the drawdown-only production model this reduces to designating the cluster with the deeper (more negative) mean drawdown as panic. The specific windows and sign assignments are listed in Appendix I. The HMM itself never sees these windows during training; they enter only at the post-estimation naming step.

The naming has no effect on the regime estimation itself. The two clusters and their boundaries are identical regardless of which is called calm or panic. What the naming controls is the interpretation of the subsequent analysis, which state we treat as the stress regime when reading off mechanism descriptions, conditional Sharpes, and cluster characterizations downstream. The label "panic" therefore reflects a statistical resemblance to historical crisis periods rather than a directly observed market state.

Filtered Regime Probability

Using the posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ from Section 3.1.5, the forward recursion (Equation 7) is applied to the full sample to obtain:

$$\pi_t^{\text{filter}} = P(s_t = \text{panic} \mid \mathbf{z}_{1:t};\, \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{P}}). \tag{12}$$

A value near 0 indicates high confidence in calm, near 1 indicates panic, and intermediate values reflect genuine ambiguity. This filtered probability serves as the regime signal passed to the cross-sectional model in Section 3.2.1.

3.2 Stage 2: Regime-Conditioned Portfolio Construction

The filtered regime probability π_t^{filter} from Stage 1 provides a monthly measure of market stress, but on its own it does not select stocks. This stage translates the regime signal into a cross-sectional portfolio. The production model is an XGBoost gradient-boosted tree regressor (XGB) that combines π_t^{filter} with momentum signals at twelve horizons to produce a monthly score for every stock. Each month, every stock in the point-in-time universe of the 1,000 largest by market capitalization is scored, and the top-scoring decile forms a long-only, value-weighted portfolio, held against the value-weighted top-1000 as the benchmark. The stock-level inputs (momentum horizons and the regime signal) are described in Section 4.

One distinction from standard momentum strategies matters here. In the traditional approach (Jegadeesh and Titman, 1993), stocks are ranked by their raw trailing return at a fixed lookback, so the top decile always holds the highest-momentum stocks. Here, stocks are ranked by their predicted forward return, a learned function of the regime signal and the full momentum term structure, so the long portfolio need not contain high-momentum stocks at all. In panic, it can hold stocks below the cross-sectional average at every horizon (Section 5.2.2).

We evaluate the strategy in long-only construction, held against the value-weighted top-1000 benchmark, for two reasons. First, the deliverable is a large-cap, fully-invested mandate: the practitioner question is whether the regime signal improves an unlevered long book relative to the index it is measured against, not whether a dollar-neutral spread earns a premium. Second, restricting to the top-1000 and dropping the short leg removes the small-capitalization and short-side frictions that dominate long-short momentum results, isolating the regime-timing contribution on the tradable long side. Performance is therefore reported as active return, tracking error, and the information ratio against the benchmark rather than as a standalone Sharpe on a self-financing spread.

3.2.1 Cross-Sectional Stock Selection Model

An XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016) is trained to predict the forward monthly return directly. Predicting the return rather than a binary above/below-median label preserves magnitude information: a stock expected to gain 10% ranks higher than one expected to gain 2%. The feature vector for each stock-month observation comprises the 13 features described in Section 4.4: the 12 momentum lookbacks ($mom_1, \ldots, mom_{12}$) and the regime signal (π_t^{filter}). By including all horizons jointly, XGBoost can learn how the full momentum term structure interacts with the regime signal, for example by conditioning its joint use of all 12 horizons on the level of π_t^{filter} in ways that cannot be decomposed into per-horizon slope changes.

XGBoost builds an additive ensemble of M regression trees, where each successive tree fits the residuals of the current ensemble:

$$F(\mathbf{x}) = \sum_{m=1}^{M} \eta \cdot h_m(\mathbf{x}), \tag{13}$$

where h_m is the m-th decision tree and η is the learning rate that shrinks each tree's contribution to prevent overfitting. Each tree h_m is grown to minimize a regularized objective combining the squared-error loss on the current residuals with penalty terms on tree complexity (number of leaves and leaf weights). The predicted return $\hat{r}^s_{t+1} = F(\mathbf{x}^s_t)$ serves as the stock score. The model is trained with $M = 500$ trees, a learning rate of $\eta = 0.05$, maximum tree depth of 4, and row and column subsampling rates of 0.8. To reduce sensitivity to random initialization, an ensemble of 20 independent XGBoost models (each with a different random seed) is trained. Their predictions are averaged before ranking stocks into portfolio deciles.

Among the nonlinear architectures flagged in Section 2.4 (gradient-boosted trees, random forests, neural networks), XGBoost is preferred because its sequential residual fitting concentrates splits on whichever feature most reduces error, turning the regime signal into the model's primary gating variable rather than one feature among thirteen.

The maximum depth sets the order of feature interactions the model can represent. We fix it at four, the smallest order at which the regime signal can condition the joint configuration of several momentum horizons: depth 1 for the regime gate plus depths 2 to 4 for a three-horizon term-structure read, rather than an interaction with single horizons in isolation. Out-of-sample performance rises to a peak at depth 4 and declines for deeper trees, whereas training cross-validation cannot resolve depth: the validation Sharpe is flat within fold noise across depths, and although the single best-tuned configuration falls at depth 3, that lead is negligible against the noise (Appendix G.4). Depth 4 is adopted for that interaction-order reason; the out-of-sample peak corroborates it but, measured on the evaluation window, is not itself the selection criterion (Appendix G.4). This aligns the model's capacity with the mechanism examined in Section 5.2.2.

With the two-stage framework in place, the next section describes the data and sample construction, and develops the evidence, from the literature and from a train-only selection exercise, for market drawdown as the HMM's sole input feature.

4 Data

This chapter describes the stock-level sample and chooses the input for the regime classifier. The production HMM observes a single series: the market drawdown. Section 4.1 develops the case for conditioning on drawdown alone, grounded in the momentum-crash literature and confirmed by a selection rule restricted to training information; Section 4.2 defines the feature and explains what the HMM adds on top of it.

The stock-level sample comprises all ordinary common shares listed on the NYSE, AMEX, or NASDAQ with a price above $1, from which a point-in-time universe of the 1,000 largest by month-end market capitalization is formed anew each month (a Russell 1000 proxy).[9] Following Shumway (1997), delisting returns are incorporated to prevent survivorship bias: actual delisting returns are used when available, and −30% is imputed for performance-related delistings with missing returns.[10] Both the regime model and the cross-sectional model are estimated on an expanding window: for each trading year they use only data through the prior December, and the out-of-sample walk runs from January 2011 to November 2025 (179 months). Table 1 provides an overview.

	Train (1990–2010)	Test (2011–2024)	Full
Unique stocks	13,363	7,037	16,657
Stock-month observations	1,161,014	557,087	1,718,101
Avg. stocks per month	4,838	3,336	4,221
Market-level months	240	167	407

Table 1: Sample summary. The sample includes all ordinary common shares listed on NYSE, AMEX, or NASDAQ with price above $1.

4.1 Choosing the Regime Indicator

The regime signal exists to answer a single question for the cross-sectional model: is the market currently in the state in which momentum's cross-sectional structure reorganizes? The choice of the HMM's observation series therefore determines which economic state the model can condition on. The production specification observes one series only: the market drawdown (DD), the percentage decline of the market index from its rolling 12-month peak, formally defined in Section 4.2. This subsection develops the case for conditioning on drawdown alone. The momentum literature identifies the market's decline-and-recovery state, rather than market stress in general, as the conditioning information that matters for momentum, and drawdown is the most direct real-time measure of that state. Statistically, a univariate HMM extracts the state with the fewest estimated parame-

[9] The $1 floor follows Gu et al. (2020); value-weighting within the top-1000 keeps the realized portfolio in liquid, large-capitalization names.

[10] Full data sources and detailed variable definitions are provided in Appendix A.

ters. Finally, a mechanical rule that uses training information only, applied to a pool of drawdown-anchored candidates, adds no further stress feature to drawdown in any year of the evaluation window.

The crash literature points to the decline state

The momentum-crash literature ties the failure mode of momentum to one specific market condition: the aftermath of a large market decline. Daniel and Moskowitz (2016) document that momentum crashes are concentrated in months that follow bear markets, identified ex ante by a negative trailing two-year market return, and attribute the crash to the option-like payoff that the loser leg acquires after severe declines: past losers rebound violently when the market turns. Cooper et al. (2004) establish the same conditioning at lower frequency: mean momentum profits are positive following UP markets and vanish following DOWN markets, where the market state is the sign of the trailing three-year market return. Asem and Tian (2010) refine the market-state evidence along the same dimension: momentum profits are higher when the market continues in its prior state than when it transitions, and the transitions out of DOWN markets are where the strategy fares worst. Goulding et al. (2023) sharpen the state definition by intersecting slow and fast trailing market returns into four states (Bull, Correction, Bear, Rebound) and show that momentum's losses concentrate around market turning points, the exits from Bear states. The pattern also has a theoretical footing: in the equilibrium model of Cujean and Hasler (2017), disagreement rises as economic conditions deteriorate, return predictability concentrates in bad times, and momentum crashes after sharp rebounds out of those states. In each case the conditioning variable is a transformation of trailing market-level returns, the depth or direction of the recent decline, and not volatility, cross-sectional dispersion, or credit conditions. Drawdown is the member of this family that measures the decline state most directly and is observable in real time: it is zero at a market peak, deepens through a decline, and narrows as the recovery progresses.

Why not volatility, and why not more features

The leading alternative conditions momentum on second moments: Barroso and Santa-Clara (2015) scale the long-short portfolio by its own trailing realized volatility and eliminate the worst crashes, and Moreira and Muir (2017) generalize the recipe by scaling factor exposures by realized market variance. Volatility management, however, is a sizing rule: it decides how much of an unconditionally constructed portfolio to hold, while leaving the selection rule unchanged. The mechanism exploited here operates through selection: in the panic state the model rebuilds the long leg toward stocks positioned for recovery (Section 5), so the relevant conditioning information is the state itself, not the risk level. Volatility is elevated in any turbulent period, whereas drawdown is near zero except during and after declines, so drawdown isolates the decline-and-recovery state itself rather than turbulence in general. Volatility does carry conditional information for momentum: Wang and Xu (2015) show that high trailing market volatility forecasts low momentum payoffs, especially in down markets, and that volatility and the market state complement each other, and Daniel and Moskowitz (2016) find that the bear-market state and ex ante market variance forecast momentum returns independently. Two considerations still favor the univariate drawdown specification. The panic regime's wider emission variance lets the filter treat deep, fast-moving drawdowns as more panic-like, but this is a property of the drawdown series itself, not a substitute for a realized-return-volatility feature. The decisive evidence is empirical: the train-only selection rule described below, whose candidate pool includes drawdown-volatility combinations, never elects a volatility feature alongside drawdown.

A broader stress set faces a different objection: estimation cost without commensurate information. Each additional emission feature adds a regime-specific mean and its share of a regime-specific covariance matrix, so a two-state model on four features carries 30 free parameters against a training panel in which panic months are rare, while the univariate drawdown model carries six.[11] Rare-regime parameters estimated from few observations lean on the prior and are fragile out of sample. Indicators tied to particular market structures can also degrade silently when the setting changes; breadth- and participation-based measures, for instance, behave differently inside a large-cap universe than in the full cross-section. Parsimony also disciplines the specification search itself: the fewer modeling choices made on the researcher's side, the smaller the scope for selection bias in the reported results. Harvey et al. (2016) document how large the implicit search space behind published return predictors has become and argue that new findings must clear correspondingly higher statistical hurdles; fixing a single, economically motivated conditioning variable in advance keeps the regime layer's researcher degrees of freedom to a minimum.

A train-only rule adds no feature to drawdown

The literature-based choice is confirmed by a mechanical check that uses no evaluation-window information. A pool of 25 drawdown-anchored candidate combinations is built from eight monthly stress indicators, all observable in real time at month end: market drawdown (DD), realized volatility (VOL), and log VIX (LVIX) for equity-market stress; cross-sectional return dispersion (DISP), relative market participation (REL_N), and cross-sectional return skewness (SKEW) for cross-sectional structure; and the BAA–AAA credit spread (CS) and the 10-year minus 2-year Treasury term spread (TERM) for credit and rates. The pool itself was fixed in advance using pre-2011 information only. Each candidate is evaluated by running the full two-stage pipeline on cross-validation folds whose validation windows end before the traded year, and candidates are compared on valida-

[11] For D features, each of the two states has D mean and $D(D+1)/2$ covariance parameters, and the transition matrix has two free entries: $2\,(D + D(D+1)/2) + 2$, which equals 6 for $D = 1$ and 30 for $D = 4$.

tion performance.[12] A one-standard-error rule then selects, among all combinations whose mean validation performance lies within one standard error of the best combination's, the one with the fewest features.[13] This rule elects the drawdown-only specification in every year of the 2011–2025 evaluation window. Because every candidate contains drawdown by construction, this exercise tests whether any additional stress feature earns its estimation cost on top of drawdown, not whether drawdown dominates every rival single indicator; the former is what the parsimony argument requires. The unrestricted alternative, taking the top-scoring combination each year, churns across six distinct multi-feature combinations over the same window without improving out-of-sample performance (Section 5). Selection is therefore not the engine of the results: the indicator is fixed on literature grounds, and the data-driven rule, restricted to training information, adds nothing to it.

4.2 The Drawdown Feature

Market drawdown is the percentage decline of the cumulative market price index from its rolling 12-month peak:

$$DD_t = \frac{P_t - \max(P_{t-11}, \ldots, P_t)}{\max(P_{t-11}, \ldots, P_t)}. \tag{14}$$

DD_t equals zero when the market stands at a 12-month high and becomes increasingly negative as the market falls below its recent peak. Drawdown is a stock variable rather than a flow: it stays negative for as long as the market remains below its trailing 12-month peak, so it encodes not merely that a decline occurred but that the market has not yet climbed back above its recent high. It is therefore informative about the recovery phase, the window in which Daniel and Moskowitz (2016) locate the momentum crash and in which the selection mechanism studied in this paper activates; persistence of the panic classification beyond the trailing window is supplied not by the feature but by the HMM transition matrix.

The HMM contributes what a raw drawdown threshold would not. First, the filter output is a probability rather than a binary state: π_t^{filter} moves continuously as evidence accumulates, so the cross-sectional model can respond in proportion to regime uncertainty. Second, the transition matrix imposes persistence, so a single month of partial rebound does not immediately flip the state, whereas hard-threshold classifications built on trailing-return signs, such as the UP/DOWN states of Cooper et al. (2004) or the ex ante bear indicator of Daniel and Moskowitz (2016), switch discretely at the boundary. Third, the level that separates the two states is learned from the training data rather than fixed in advance; Daniel et al. (2019), who model momentum's calm and turbulent states with

[12] The validation metric is the information ratio of the pipeline's portfolio against the value-weighted benchmark over the fold's validation months.

[13] The standard error is computed from the dispersion of the best combination's per-fold validation scores; ties at the minimum feature count are broken by the higher validation mean.

exactly this class of two-state hidden Markov model (estimated on joint market and momentum returns), explicitly fault fixed-lookback definitions of depressed markets as arbitrary, and treating the boundary as a latent-state estimation problem answers that critique. The construction formalizes the point of Hamilton (1989) that the regime is latent and must be inferred probabilistically from what is observed.

The drawdown series is winsorized at the 1st and 99th percentiles[14] using training-sample statistics, and standardized to zero mean and unit variance.

Panel (a) of Figure 3 plots the standardized series over the sample. Panic regimes are characterized by deep drawdowns that are slow to close; calm regimes show shallow or no drawdown. The series deepens sharply in the 2000–2002 dot-com bust, the 2007–2009 Global Financial Crisis, the March 2020 COVID shock, and the 2022 tightening cycle, and sits near zero through the 2003–2007 and 2013–2019 expansions. The largest excursions align with the shaded NBER recession bands, though drawdown also flags equity-market stress that falls outside official recession dating.

4.3 Regime Estimation Results

The regime model is re-estimated every year of the walk rather than once on a fixed training block: for trading year Y the sampler runs on the drawdown panel through December $Y - 1$, and the causal forward filter (Section 3.1.6) is applied through year Y. The regime signal evaluated below is therefore the concatenation of these out-of-sample yearly filters, the probability path that actually drove the portfolio, rather than a single in-sample fit.

Sampler configuration and convergence

The sampler settings are inherited from the validated thesis configuration: 2,000 Gibbs iterations with 500 discarded as burn-in, with the filtered probability averaged across independent seeds per fit (Section 3.1.5). Convergence of the multi-feature thesis specification is documented at length in Appendix J via the effective sample size and the Gelman–Rubin $\hat{R}$; the drawdown-only model estimated here is a one-dimensional restriction of that specification, so its emission space is far better identified, and label switching, the only genuine multimodality in a two-state Gaussian HMM, is resolved deterministically by assigning the deeper-drawdown cluster to the panic state (Section 3.1.6). Averaging the filtered probability across seeds cancels the residual seed-specific Monte Carlo noise.

Regime separation and the filtered signal

Panel (b) of Figure 3 plots the resulting out-of-sample filtered probability π_t^{filter} over the 2011–2025 evaluation window. The signal is sharply bimodal: its median is 0.005 and it

[14] Winsorizing at the 1st and 99th percentiles caps any value below the 1st percentile at that bound and any value above the 99th percentile at that bound, limiting the influence of outliers without discarding the observations.

exceeds the 0.5 panic threshold in 45 of the 179 months (25%), so the great majority of months are classified as unambiguously calm. Elevated-probability episodes line up with the recognized stress windows of the period, the 2011 euro-area and U.S. downgrade sell-off, the 2015–2016 China and oil shock, the late-2018 drawdown, the March 2020 COVID crash, the 2022 tightening bear market, and the April 2025 tariff episode. The 2022 bear market is the longest single panic run at fourteen consecutive months.

The regime signal separates the states on drawdown exactly as intended. Across the evaluation window the market sits, on average, 9.9% below its trailing peak in panic-classified months versus 0.7% in calm months; in standardized units the mean drawdown is -0.57 in panic months against $+0.58$ in calm months. The filtered probability rises strongly with the depth of the drawdown, with a Pearson correlation of 0.86 between π_t^{filter} and the drawdown depth, but it responds to the full standardized-drawdown history rather than the contemporaneous level alone. The transition matrix supplies the persistence visible in the figure: isolated one-month dips do not trip the signal, whereas sustained declines push it toward one and hold it there until the drawdown closes.

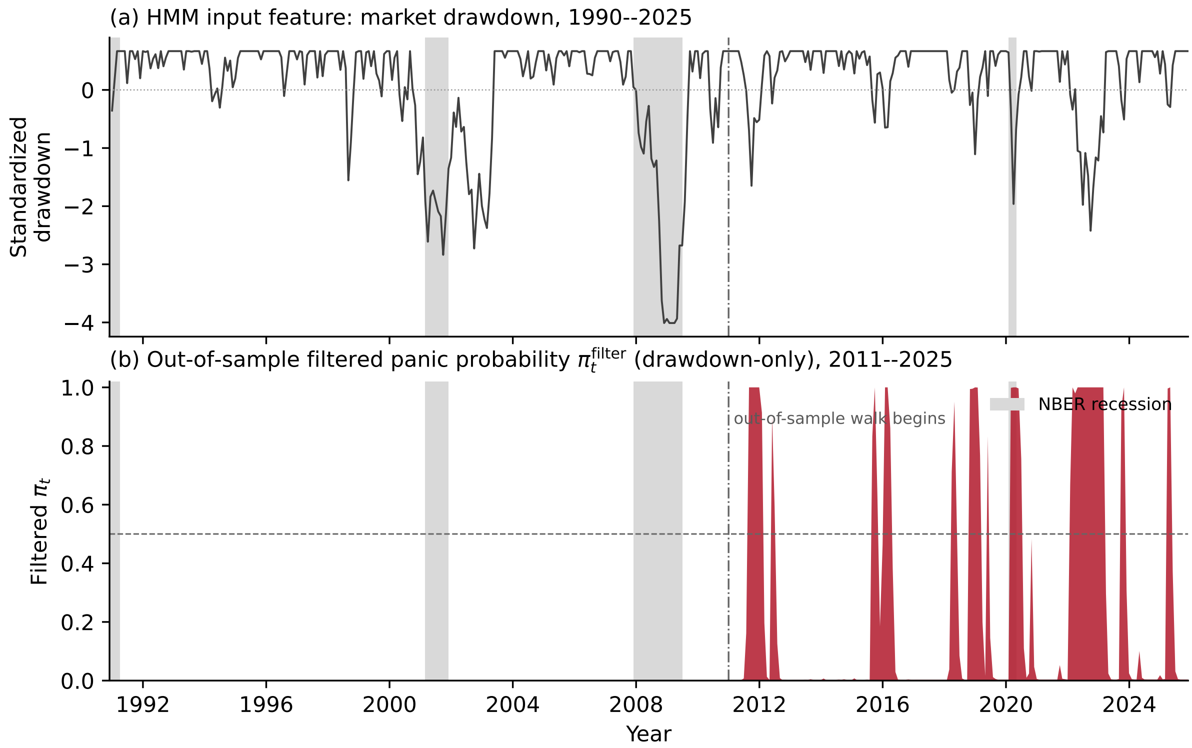

Figure 3: The drawdown regime signal. Panel (a): the standardized market drawdown, the sole HMM input feature, over 1990–2025; the dash-dotted line marks the start of the out-of-sample walk. Panel (b): the out-of-sample filtered panic probability π_t^{filter} from the drawdown-only model (the signal that drove the portfolio), 2011–2025, with the 0.5 panic threshold dashed. Grey bands mark NBER recessions.

4.4 Cross-Sectional Features

The cross-sectional model ranks stocks each month using twelve trailing momentum signals together with the filtered panic probability π_t^{filter}. The latter varies only at the monthly market level (every stock in the same month receives the same value), so it functions as a conditioning variable rather than a stock-level predictor.

Twelve trailing momentum signals are computed for each stock, corresponding to lookback horizons of 1 to 12 months:

$$\text{mom}^s_{k,t} = \prod_{\ell=1}^{k}(1 + r^s_{t-\ell}) - 1, \qquad k = 1, \ldots, 12, \tag{15}$$

where $r^s_{t-\ell}$ is the adjusted return of stock s in month $t - \ell$. The current month's return is excluded from all lookbacks to avoid mechanical correlation with the forward return being predicted. Including all twelve horizons rather than a single lookback allows the models to learn how the full momentum term structure interacts with the regime signal.

4.5 International Sample

For the robustness check in Section 5.3.4, the same pipeline is applied to UK and Japanese equities from Compustat Global over the same 1990–2024 window. The HMM feature set is re-selected per region on each market's data (Appendix K), with the candidate pool adapted to use locally available stress indicators. Both regions select *DD+VOL+REL_N*. Long and short legs are formed from unconditional decile breakpoints across all listed stocks, since no NYSE-equivalent tier exists for either market. Per-region selection details and stability checks are in Appendix K.

With the regime signal and the cross-sectional features in place, the next chapter evaluates the regime-conditioned portfolio out-of-sample and decomposes the model's monthly selections.

5 Results

This chapter answers how the cross-sectional structure of momentum changes across regimes. The XGBoost model is the main method; the deterministic formula and the logistic regression are reported as benchmarks on the same inputs. Section 5.1 compares the three methods on the 2011–2024 out-of-sample window and isolates the contribution of the regime signal through targeted ablations. Section 5.2 decomposes the XGBoost model's per-month picks, first via per-horizon contribution analysis and then via a clustering of the long-leg cross-sectional z-curves that recovers four recurring selection modes. Section 5.3 stress-tests the result with a risk-aversion sweep, a 25-year expanding-window backtest covering the dot-com bust and the GFC, and an international replication on UK and Japanese equities.

5.1 Portfolio Performance

5.1.1 Results Overview

	Ann. Ret	Ann. Vol	Sharpe	Max DD	β	NW t	Final \$
Market	11.9%	14.6%	0.84	-24.7%	1.00	–	4.8
Fixed 12-mo mom	-1.9%	26.1%	0.06	-72.2%	−0.66	0.24	0.8
Fixed 1-mo mom	3.1%	17.6%	0.26	-30.2%	−0.34	1.08	1.5
DET	-0.2%	23.6%	0.11	-63.8%	−0.59	0.43	1.0
LR	-1.3%	15.2%	-0.01	-42.4%	0.05	-0.04	0.8
OLS	-11.1%	17.4%	-0.58	-82.0%	0.27	-2.25**	0.2
XGB	21.7%	19.5%	1.11	-22.8%	0.45	4.37***	15.3

Table 2: Out-of-sample portfolio performance.

Notes: Long-short (top/bottom decile, NYSE breakpoints), value-weighted, net of 10 bps one-way costs. β is the market beta. NW t uses Newey–West standard errors (6 lags). Test period: January 2011 to November 2024 (167 months). $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

Table 2 reports out-of-sample performance[15] for the main XGBoost model (XGB) and its two same-input benchmarks (DET, LR), alongside three external benchmarks: the value-weighted U.S. market and two unconditional momentum portfolios at the dominant 12-month and contrarian 1-month horizons. The three model variants share the same inputs, the twelve monthly momentum horizons together with the HMM filtered probability π_t^{filter}, but differ in functional form: DET is a deterministic rule that interpolates between mom_1 and mom_{12} as π_t^{filter} varies; LR is a logistic regression on those features; XGB is an XGBoost regressor capable of representing arbitrary nonlinear interactions.

[15] Sharpe convention details (raw vs. excess returns, arithmetic-vs-geometric mean) are in Appendix C.3.

The pattern across rows is clear. The value-weighted market earns 11.9% annualized at Sharpe 0.84, reported as the passive equity benchmark.[16] Unconditional 12-month momentum (the canonical anomaly) delivers only Sharpe 0.06 with a −72.2% maximum drawdown, exposing the crash vulnerability documented by Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015). Of the regime-conditional variants, the deterministic rule (DET) improves on 12-month momentum only marginally to 0.11, and the logistic regression (LR), given identical features to XGB, collapses to −0.01. Only XGB delivers economically meaningful performance, an annualized Sharpe of 1.11 with a six-factor alpha of 24.1% ($t = 4.81$; Appendix H.3).[17] It is the only row in Table 2 whose Newey–West t-statistic clears every conventional threshold. Block-bootstrap inference confirms this gap is not driven by sampling noise (Appendix H.4).

The Newey–West truncation of six lags is empirically supported by Ljung–Box diagnostics on the factor-regression residuals. For the headline XGB six-factor regression, LB(6) = 10.74 ($p = 0.097$) and LB(12) = 17.51 ($p = 0.132$); the test supports the 6-lag truncation for every (strategy, model) pair ($p > 0.05$ throughout). The full diagnostic table and the residual ACF figure appear in Appendix H.3.

A further specification confirms that the limiting factor is linearity itself. Hand-engineering the missing nonlinearity into LR through π^2 and $\pi \times mom_k$ terms lifts it to 0.32, establishing that the regime-momentum interaction is real while also showing that pre-specified interactions recover less than a third of what XGBoost finds automatically. Adding firm-level fundamentals to XGB's feature set degrades performance rather than improving it (Section 5.1.3), evidence that π_t^{filter}'s role is specifically to condition the momentum term structure. XGB is therefore adopted as the production model for the remainder of the thesis.

The regime-conditional view (Table 3) shows where this advantage comes from. In panic, exactly where unconditional momentum is known to crash, XGB earns a Sharpe of 1.53. In calm, where the conventional continuation signal is strongest, it earns 0.84. The panic figure is roughly double the calm one, consistent with the regime-conditioning pattern the architecture is designed to produce.[18]

[16] The market is long-only ($\beta = 1$), so its raw Sharpe is not directly comparable to the long-short rows in Table 2; the model variants' like-for-like peers are the unconditional long-short momentum benchmarks.

[17] The factor alpha is the intercept of a regression of long-short returns on the market, SMB, HML, RMW, CMA, and UMD factors. Including UMD is the toughest version of the test for a momentum strategy: the regression cancels any return earned by mechanically tilting toward winners. Full specification in Appendix H.3.

[18] The calm-panic gap is not statistically significant ($p = 0.109$ by block-bootstrap on 59 panic months; Appendix H.4), but every benchmark still trails XGB widely in panic (LR: 0.39, Fixed 12-mo: 0.05, vs XGB's 1.53).

	Full	Calm	Panic
Market	0.84	0.64	1.13
Fixed 12-mo mom	0.06	0.08	0.05
Fixed 1-mo mom	0.26	0.52	−0.01
DET	0.11	0.18	−0.01
LR	−0.01	−0.29	0.39
XGB	1.11	0.84	1.53

Table 3: Regime-conditional Sharpe ratios (108 calm months, 59 panic months).

Figure 4 visualizes the same pattern through cumulative wealth paths from January 2011. XGB's advantage concentrates in the shaded panic months, motivating the regime-by-regime decomposition that follows.

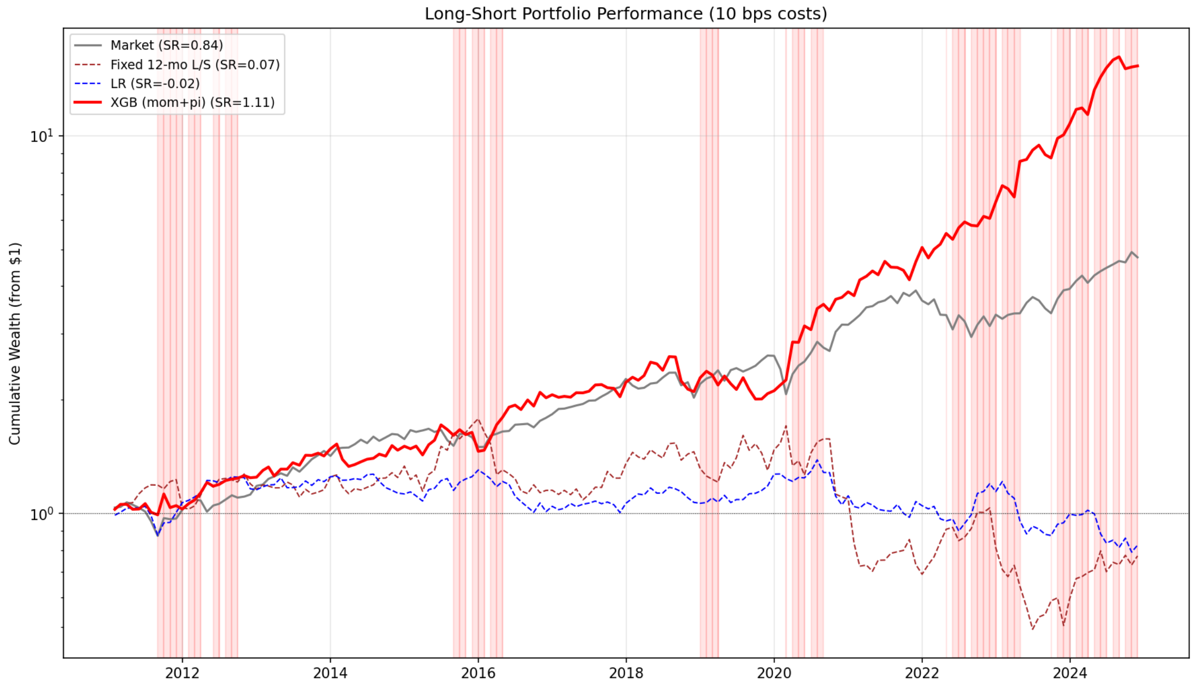

Figure 4: Cumulative wealth paths (log scale) starting from $1 in January 2011. Shaded bands indicate panic months ($\pi_t^{\text{filter}} \geq 0.5$).

5.1.2 Regime Signal Ablation

The regime signal is essential. Removing π_t^{filter} and retraining on the 12 momentum features alone drops the Sharpe to 0.43 (Table 4). Replacing π_t^{filter} with the four raw stress indicators (DD, DISP, REL_N, CS) performs even worse (Sharpe 0.26). With four additional dimensions the model must reconstruct the regime structure at each split, while the HMM distills those indicators into a single dimension that one tree split can condition on. The HMM's role is not to provide a convenient feature but to solve regime identification *before* stock selection, freeing the cross-sectional model to use the regime signal rather than construct it.

Furthermore, reducing the raw stress indicators to a single feature does not match the HMM. Across the four HMM inputs tested singly, the best (relative market participation, REL_N) gives Sharpe 0.75 and the conventional drawdown indicator (DD) gives 0.47, neither approaching the production XGB Sharpe of 1.11. The simple regime classifiers used in the asset-pricing literature, such as the up/down state of Cooper et al. (2004) from trailing market returns or single-threshold drawdown rules, are essentially raw single-feature signals of this kind. The HMM is performing regime classification, a task XGB cannot perform from raw stress data, and the added cost of the upstream HMM stage is therefore justified rather than optional.

Regime Signal Variant	Ann. Ret	Ann. Vol	Sharpe	Max DD
XGB + π_t^{filter} (HMM)	21.7%	19.5%	1.107	−22.8%
XGB + REL_N only	11.6%	16.6%	0.746	−24.5%
XGB + DD only	7.9%	20.5%	0.469	−30.8%
XGB (no regime signal)	7.0%	20.4%	0.429	−41.0%
XGB + raw indicators (DD, DISP, REL_N, CS)	3.2%	19.1%	0.257	−45.4%

Table 4: Regime signal ablation (50-seed ensemble). The HMM-distilled π_t^{filter} outperforms any raw stress feature alone, the four features concatenated, and the no-signal baseline. The two single-feature rows show the strongest (REL_N) and the conventional drawdown indicator (DD); the remaining single-feature variants (DISP = 0.17, CS = 0.02) sit at or below the no-signal baseline.

5.1.3 Fundamental Feature Augmentation

A complementary question is whether *adding* widely used firm-level fundamentals to the production feature set strengthens the strategy. Ten characteristics are tested,[19] each trained at production settings.

	Feat.	Sharpe	Calm	Panic	Fund SHAP
Baseline (mom + π)	13	1.11	0.84	1.53	0%
Baseline + 10 fundamentals	23	0.89	0.69	1.18	44%
Fundamentals only	10	0.59	0.86	0.38	100%
Fundamentals + π	11	0.64	0.59	0.72	60%
Momentum + fund. (no π)	22	0.63	0.68	0.57	54%

Table 5: Feature-set ablation (50-seed XGBoost ensembles, production hyperparameters). Full Sharpe is reported alongside calm-period and panic-period Sharpe. "Fund SHAP" is the share of total SHAP attention absorbed by the ten fundamental features.

[19] Book-to-market, return on equity, gross profitability, asset growth, leverage, earnings growth, log market equity, operating cash flow over assets, free cash flow over assets, and accruals. All real-time and four-month-lagged from Compustat `fundq`, with full definitions in Appendix L.

Adding the ten fundamentals on top of the production feature set reduces Sharpe from 1.11 to 0.89 (Table 5), with the loss concentrated in panic months: panic Sharpe falls from 1.53 to 1.18 while calm Sharpe drops only from 0.84 to 0.69. The fundamentals absorb 44% of total SHAP, redirecting attention from the momentum term structure (54% to 27%) and the regime signal (46% to 29%).

Fundamentals describe firm-level characteristics whose cross-sectional ranking does not flip with the market regime the way the optimal momentum tilt does. A high-B/M stock is a high-B/M stock in both regimes, so fundamentals cannot reproduce the panic-side reversal that drives the baseline's panic Sharpe (Table 5, fund-only and fund $+\pi$ rows). Adding them on top of the regime-momentum signal therefore dilutes an essential mechanism rather than complementing it. A regime-conditional feature-set design that routes fundamentals into calm-regime predictions and momentum into panic-regime predictions is discussed in Section 6.3.

5.2 The Regime-Momentum Mechanism

This section examines the XGBoost model's internals, reading the structure of the regime-momentum signal directly off its per-month selections.

We first analyze the contribution by horizon (Section 5.2.1), then dig deeper into the common decision patterns the model produces (Section 5.2.2), and finally discuss the results (Section 5.2.3).

5.2.1 Momentum Contribution by Horizon

XGB's predictions are decomposed using SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017), which measure how much each input contributes to the model's prediction (definitions in Appendix D.1). A single feature, π_t^{filter}, carries 46% of total SHAP, with the 12 momentum horizons together accounting for the other 54% (Table 6). Equal weighting would give π_t^{filter} just $1/13 \approx 8\%$ of total SHAP, so the model loads on the regime signal roughly six times more than its share of the feature set. The concentration on a single feature supports a routing-variable role: π_t^{filter} determines how the model uses the momentum term structure. Between regimes, momentum's share rises from 51% in calm to 59% in panic while π_t^{filter}'s share falls from 49% to 41%. The regime signal does most of its work where the regime call is ambiguous (calm and the calm/panic boundary) and recedes once π_t^{filter} saturates near 1 in deep panic.

Feature	Overall	Calm	Panic
Momentum (12 horizons)	54%	51%	59%
π^{filter}	46%	49%	41%

Table 6: SHAP feature importance shares for XGB. Each column sums to 100%.

The aggregate split is reported by leg and horizon in Figure 5. Both legs concentrate on months 8–12 (about 65% each), with months 11 and 12 carrying the largest individual shares and the short leg's peak at month 11 (22%) the most concentrated single-horizon share in the figure. This is consistent with the classical Jegadeesh and Titman (1993) 12-month momentum range, where the cross-section's momentum signal is historically strongest, whether the model is identifying winners or losers.

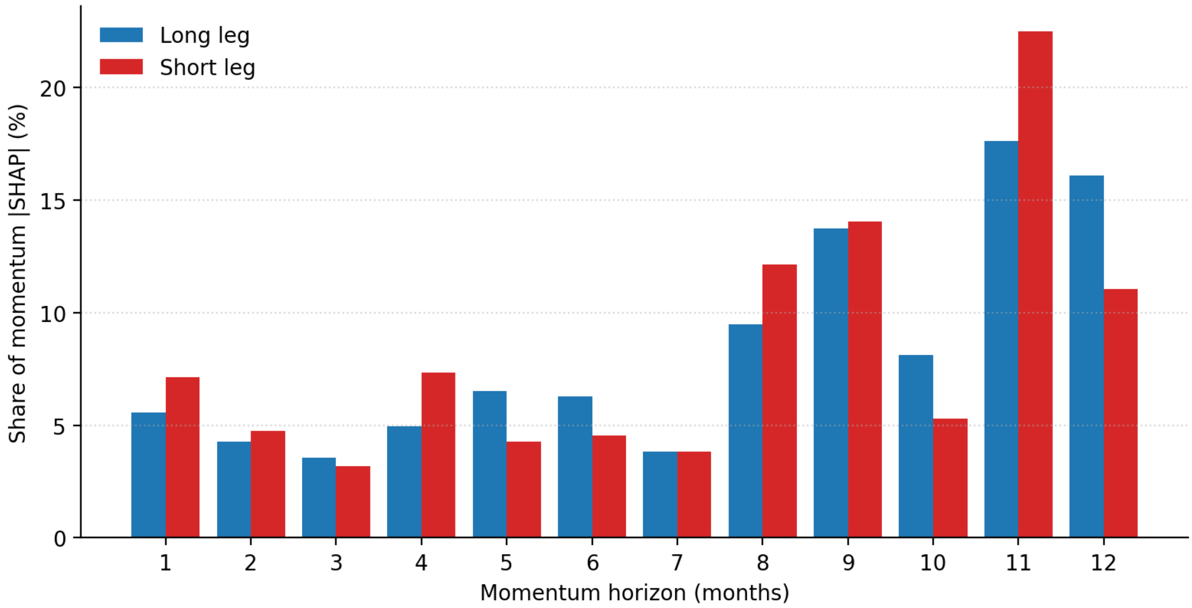

Figure 5: Per-horizon SHAP shares of momentum features in XGBoost's predictions, by leg, pooled across calm and panic regimes weighted by stock-month counts.

SHAP shares show which horizons matter, but not in which direction: the same SHAP weight could reflect winner-selection or loser-selection at that horizon. Whether XGB is treating the momentum term structure as a continuation signal, a reversal signal, or something that flips with the regime is a question the aggregate SHAP table cannot answer. Section 5.2.2 resolves this by tracing the picks themselves through their cross-sectional z-scores.

5.2.2 Cross-Sectional Z-Scores

To trace the picks themselves, we look at the cross-sectional z-scores of the stocks XGB selects for the long leg. For each stock s the model has placed in the long leg, the z-score at horizon k is $(mom_k^s - m\bar{o}m_k)/\sigma_k$, where $m\bar{o}m_k$ and σ_k are the cross-sectional mean and standard deviation of horizon-k trailing momentum across all stocks in the universe that month. The long-leg z-curve for a given month is the average of these per-stock z-scores across all stocks the model selected for the long leg, giving a 12-element profile describing where the model's picks sit relative to the cross-sectional mean at each lookback. The sign indicates which side of the cross-sectional mean the basket sits on at horizon k (positive:

above the mean; negative: below). The magnitude measures how far it sits in standard-deviation units: a z-score of +0.8 means the model's picks have, on average, trailing momentum at horizon k that is 0.8 cross-sectional SDs above the average that month, a more pronounced tilt than +0.2. Figure 6 plots each month's long-leg z-curve as one row.

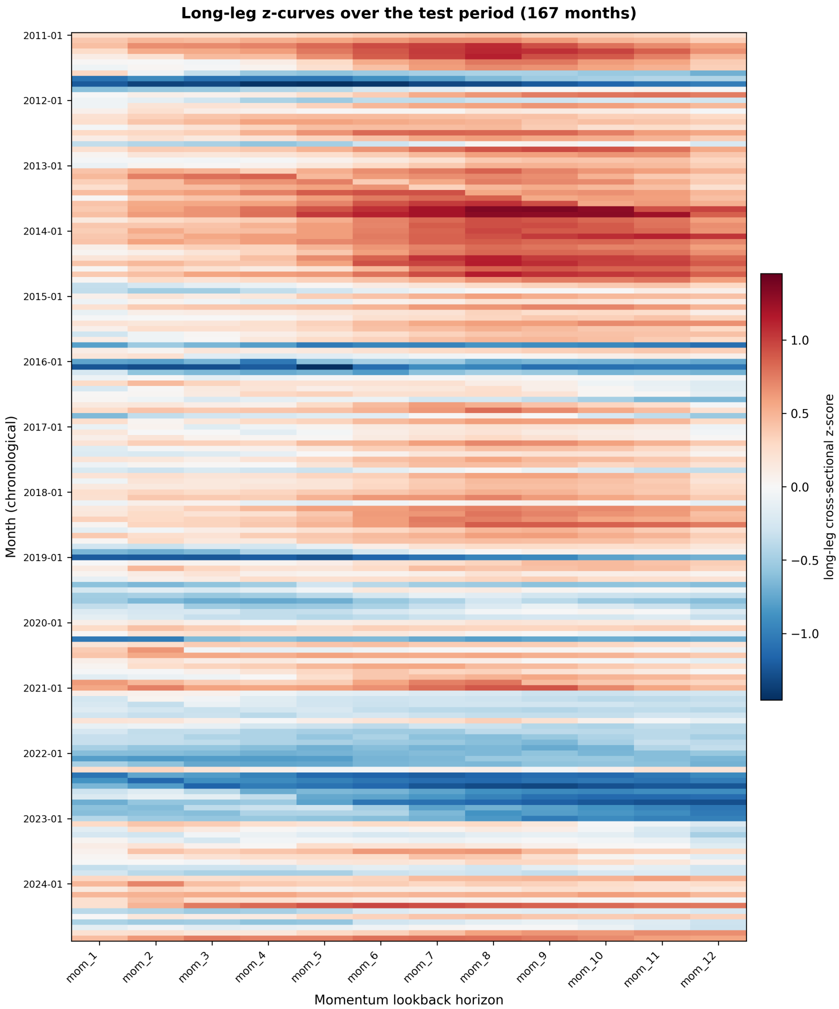

Figure 6: Long-leg cross-sectional z-scores by month and momentum horizon, 2011–2024. Each row is one of 167 test months sorted chronologically. Each column is one of the 12 momentum horizons. Color shows the deviation of the long leg's mean momentum at horizon h from the cross-sectional mean trailing momentum that month.

The heatmap reveals a small number of recurring selection modes. To summarize these, we cluster the 167 monthly z-curves with K-means using Euclidean (L_2) distance. Each month's z-curve is a point in 12-dimensional space, one coordinate per momentum horizon. The L_2 distance between two months A and B is the square root of the sum of squared per-horizon z-score differences, $\sqrt{\sum_{k=1}^{12}(z_k^A - z_k^B)^2}$, so two months are close when their long-leg baskets sit at similar z-scores at each of the twelve horizons. K-means partitions the 167 months into $K = 4$ groups by minimizing total within-cluster L_2 distance to each group's centroid. $K = 4$ is chosen on economic grounds. We do not use $K = 3$ because the coarser partition collapses the two below-average reversal modes (post-panic recovery and deep-crisis term-structure-wide reversal) into a single cluster, eliminating a distinction central to the mechanism (Section 5.2.3). We do not use $K \geq 5$ because the four-cluster partition already covers the two above-average modes and two below-average modes that the mechanism turns on, and further splitting would partition an economically homogeneous group without revealing a new selection pattern.

The partition is stable across random initializations: 97.6% of months get the same label across six seeds, with mean pairwise Adjusted Rand Index 0.96. The four groups nest inside a coarser two-fold partition. Clusters 1 and 2 are the 115 months in which the long leg sits above the cross-sectional mean ("above-average baskets", $\bar{\pi} = 0.28$). Clusters 3 and 4 are the 52 months below ("below-average baskets", $\bar{\pi} = 0.52$).

Per-cluster descriptive statistics are in Table 7. The full feature signature is in Appendix Table 26. Each cluster's z-curve panel appears immediately above its description (Figures 7–10). Thin lines are member-month z-curves and the bold line is the cluster centroid.

Throughout the cluster descriptions below, *cross-sectional* statistics are averaged across the full stock universe each month and *long-leg* statistics across the model's selected names. The contrast between the two is the core diagnostic. Each cluster's Sharpe is computed from the realized long-short monthly returns over the months assigned to that cluster: $\text{Sharpe}_C = \text{mean}(r_{M_C})/\text{std}(r_{M_C}) \cdot \sqrt{12}$, where M_C is the set of months in cluster C. Confidence intervals come from a block bootstrap (block size 6, 5,000 reps).

Feature	1 (calm)	2 (transition)	3 (post-panic)	4 (deep crisis)
Months (n)	53	62	31	21
$\bar{\pi}_t^{\text{filter}}$ (current panic prob.)	0.21	0.34	0.32	0.81
Cross-section std of 9–12 mo momentum	0.60	0.59	0.85	0.45
Cross-section skewness of 1–4 mo momentum	4.6	7.2	6.4	3.7
Long-leg avg z-score $\bar{z}$	+0.57	+0.19	−0.29	−0.82
Long-leg avg trailing return	+40%	+16%	−3%	−37%
Cross-section avg trailing return	+14%	+7%	+13%	−9%
Short-leg avg trailing return	−3%	+0%	+42%	−4%
Cluster Sharpe	0.92	0.93	1.21	1.82

Table 7: $K = 4$ cluster descriptors, 2011–2024 ($n = 167$ months); columns are clusters. The trailing-return rows (long, cross-section, short) are averaged across the 12 momentum horizons. Dispersion is shown at 9–12 months and skewness at 1–4 months, the windows where clusters differ most. Cluster Sharpe is annualised, block-bootstrapped (block size 6, 5,000 reps).

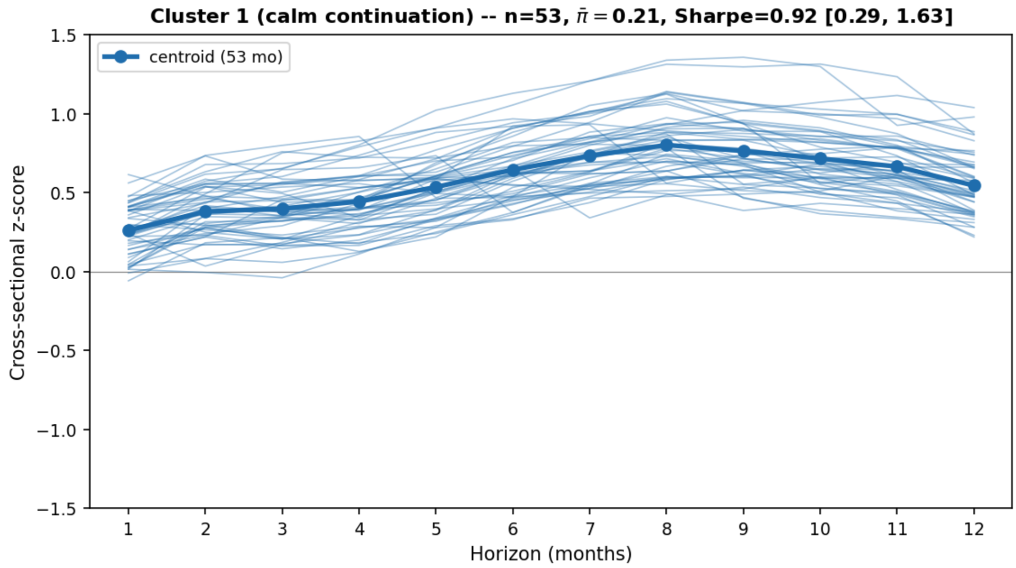

Figure 7: Cluster 1 (calm continuation): z-curves of the 53 member months and centroid. The centroid is humped and uniformly positive, peaking at +0.80 at month 8.

Cluster 1 (calm continuation): a quiet, broadly trending market

The long-leg z-curve has a z-mean of +0.57, the largest average amplitude on the continuation side, with a humped shape. Picks sit only mildly above average at short horizons ($z \approx +0.26$ at month 1) but are well differentiated at intermediate horizons (peak +0.80 at month 8). In absolute terms, the long-leg mean trailing return reaches +60% at the 9-month horizon. The cluster is concentrated in 2013–2014, running in stretches of up to 21 consecutive months.

These 53 months are a stable, mature bull market. With cluster-mean $\bar{\pi}_t^{\text{filter}} = 0.21$ (the lowest of any cluster), the model has high confidence the regime is calm and commits fully to its standard winner-selection mode. The cross-sectional mean trailing return is positive at every horizon (the average stock returned +3%/+13%/+23% over 1/6/12 months), so there is a coherent up-trend for the model to exploit. The rise is broad rather than driven by a few outliers (cross-sectional return skewness 4.6/6.2/6.7, below the test-period average of 5.8/6.6/7.1), which means winner-selection picks up genuine widespread participation rather than chasing a handful of unrepresentative names.

This is the canonical setting for cross-sectional momentum, the kind of extended bull market in which Cooper et al. (2004) document that momentum profits are most concentrated. The picks reproduce the buy-side of Jegadeesh and Titman (1993)'s twelve-month relative-strength strategy, with the intermediate-horizon peak matching Novy-Marx (2012)'s documented dominance. The peak is where Hong and Stein (1999)'s gradual information diffusion predicts the strongest signal: news enters prices slowly, so winners keep winning at intermediate horizons until the information is fully absorbed. The hump's low end at month 1 is consistent with the low SHAP share month 1 receives in Section 5.2.1. It is also consistent with the short-horizon reversal and microstructure noise long documented in the literature (Jegadeesh, 1990; Lehmann, 1990).

Cluster 1 earns a Sharpe of 0.92, the lowest of the four. Continuation still pays, but the strategy is following an established rally rather than catching a turning point. Cluster 1 serves as a sanity check on the architecture: at the lowest π_t^{filter} values in the test sample the model recovers the classical cross-sectional momentum pattern, so the regime-conditioning architecture adds new behavior during stress without overriding the unconditional pattern that already works in calm.

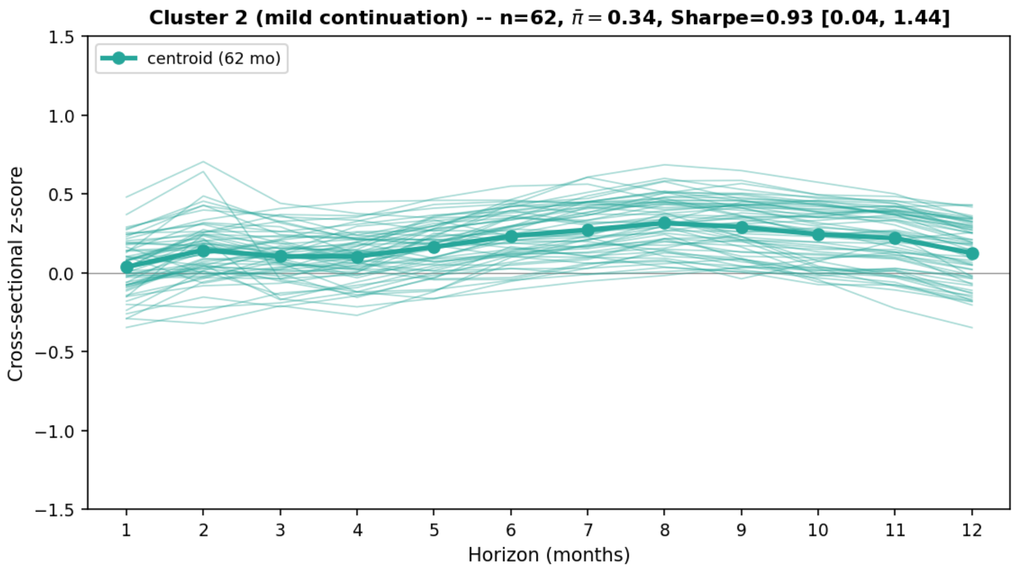

Figure 8: Cluster 2 (transition): z-curves of the 62 member months and centroid. The centroid is mildly positive throughout, peaking at +0.32 at month 8.

Cluster 2 (transition): a choppy market with no broad direction

The long-leg z-curve is nearly flat, with most horizons sitting between +0.10 and +0.30 (z-mean +0.19, peak +0.32 at month 8), the smallest absolute amplitude of any cluster, reflecting a hedged selection rather than a confident commitment. Cluster 2 is the largest of the four ($n = 62$) but appears only in short bursts (max 4 consecutive months) spread throughout the test period.

These 62 months are an ambiguous transition state, with cluster-mean $\bar{\pi}_t^{\text{filter}}$ in a middle band (0.34, neither clearly calm nor clearly panic). The cross-sectional average momentum is mild ($mom_{\text{overall}} = 0.07$, below the test-period average of 0.08), so no clear up-trend has formed. The cross-sectional mean trailing return is only mildly positive (the average stock returned +2%/+8%/+11% over 1/6/12 months) and cross-sectional return skewness is the highest of any cluster (7.2/7.6/8.1), with a small number of extreme winners pulling the mean above zero while the median stock's return stays near flat, so even the modest positive level is not broad-based. With π_t^{filter} between calm and panic without decisive separation, the model plays safe and earns its return through occasional outlier capture rather than systematic selection. The short-burst pattern is consistent with cluster 2's role as a brief transition between regimes rather than a stable state. The muted shape is consistent with Cooper et al. (2004)'s finding that momentum profits fade when the market state is ambiguous, and with Hong et al. (2000)'s slow-diffusion mechanism, in which profits concentrate in stocks completing their delayed information adjustment rather than spreading across the cross-section.

Cluster 2 earns a Sharpe of 0.93, essentially identical to cluster 1's 0.92 but reflecting a different mechanism. Where cluster 1 exploits a confidently-tilted winner basket, cluster 2 picks up its return through occasional right-tail outliers. The cluster exposes the limit of the regime-momentum architecture when π_t^{filter} sits in the ambiguous middle and the cross-section carries no clear directional trend.

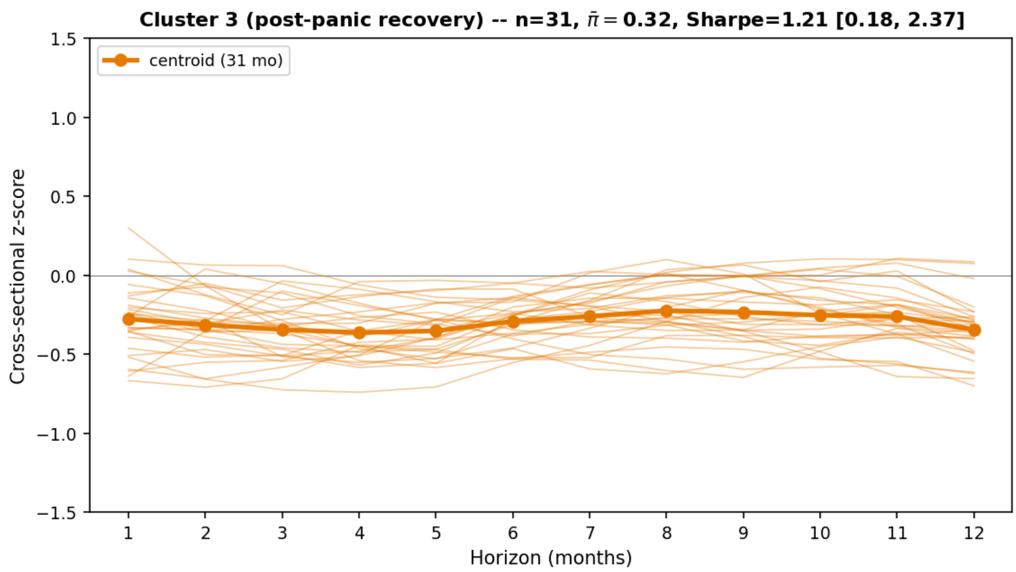

Figure 9: Cluster 3 (post-panic recovery): z-curves of the 31 member months and centroid. The centroid sits below the cross-sectional mean at every horizon, between −0.22 and −0.36.

Cluster 3 (post-panic recovery): a recovering market with high long-horizon dispersion

The long-leg z-curve sits below the cross-sectional mean at every horizon, with z-mean −0.29. The deepest below-average points are at month 4 (−0.36) and month 12 (−0.35). The amplitude is moderate, smaller than the extreme commitments seen in clusters 1 and 4. The long-leg mean trailing return ranges from −6% to +1% across horizons, while the cross-sectional mean trailing return is up double digits. Cluster 3 is concentrated in 2021 (9 of 31 months, the post-COVID rotation) and 2023.

These 31 months sit at a moderate cluster-mean panic probability ($\bar{\pi}_t^{\text{filter}} = 0.32$) paired with a strongly recovering cross-sectional mean trailing return (the average stock returned +1%/+11%/+27% over 1/6/12 months). The strong 12-month return paired with a near-flat most recent month implies the rally is mature. The bulk of the rebound has already happened in the prior 6–12 months, and cluster 3 catches the late-stage decelerating phase. The joint signature of moderate π and strong positive cross-sectional average momentum conditions the trees to select from a post-bear-rebound branch, where the long leg is built from cross-sectional losers rather than winners. Cross-sectional return dispersion is the highest of any cluster at both the 5–8 and 9–12 month horizons (0.56 and 0.85), reflecting an uneven recovery in which some stocks rebound while others still lag. The increasing dispersion across horizons matches Lee and Swaminathan (2000)'s lifecycle interpretation, in which momentum signals evolve from continuation to reversal as information ages.

These months also exhibit the beta-reversion dynamic that Daniel and Moskowitz (2016) identify as crash-prone for unconditional momentum. During the preceding bear market, high-β stocks fell disproportionately and low-β defensives held up. When recovery arrives, the high-β names rally disproportionately and the defensives lag. Fixed 12-month momentum therefore enters the recovery on the wrong side of the trade, long the defensives and short the high-β rebounders, and loses money. Conditioning on π_t^{filter} reverses XGB's picks: it puts the high-β rebounders in its long leg and the defensives in its short leg, so the same rally that crashes unconditional momentum lifts XGB instead. Cluster-level CAPM betas confirm this (Section 5.2.3 below and Appendix H.7, Table 27).

Cluster 3 earns a Sharpe of 1.21, in the same months where classical momentum is most vulnerable. What is distinctive is the counter-intuitive bet that selecting stocks still below average in a recovering market is more profitable than chasing the names that have already recovered.

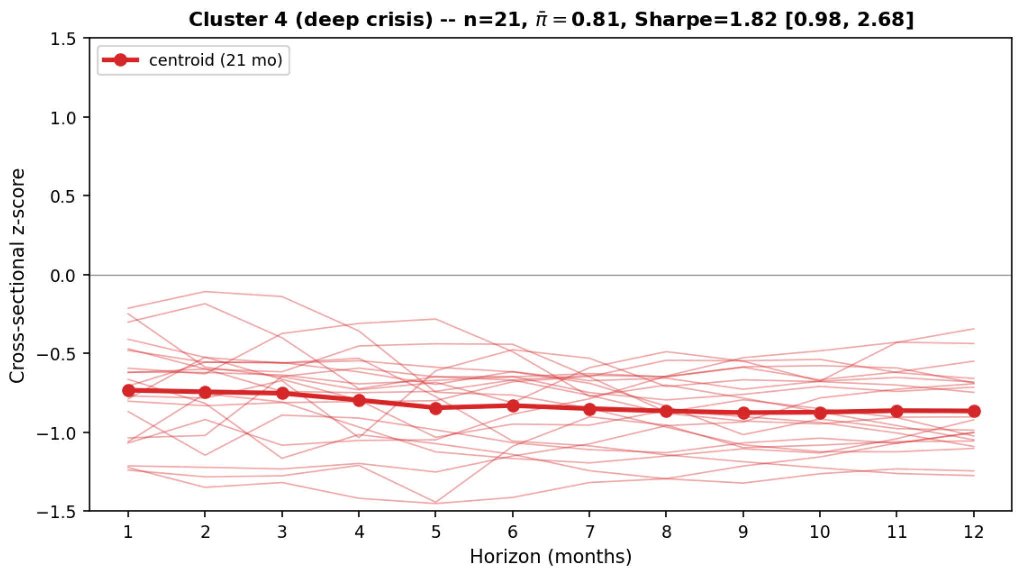

Figure 10: Cluster 4 (deep crisis): z-curves of the 21 member months and centroid. The centroid sits uniformly between −0.73 and −0.88 across all 12 horizons.

Cluster 4 (deep crisis): an active panic with the whole market falling

The long-leg z-curve sits uniformly between -0.73 and -0.88 across all 12 horizons (deepest around month 9, z-mean -0.82), meaning the picks are roughly 0.7 to 0.9 cross-sectional standard deviations below the cross-sectional mean at every horizon. The long-leg mean trailing return deepens from -15% over the prior month to -48% over the prior 12 months. Eleven of 21 months are in 2022 (Fed-hiking cycle), with corroborating instances in April 2020 (COVID trough), October 2015 and January 2016 (China devaluation, oil collapse), and January 2023.

The market is in active panic, with cluster-mean $\bar{\pi}_t^{\text{filter}} = 0.81$ (the highest of any cluster) and the cross-sectional mean trailing return uniformly negative (the average stock returned $-5\%/-11\%/-7\%$ over 1/6/12 months). The 6-month return is more negative than the 12-month return, so the older half of the year was roughly flat or positive and the steep decline is concentrated in the most recent months, the signature of a short panic rather than a prolonged year-long deterioration. Cross-sectional return dispersion is the lowest of any cluster at both the 5–8 and 9–12 month horizons (0.35 and 0.45), and 1–4 month cross-sectional return skewness is also the lowest (3.67), so stocks are moving down together rather than fanning out.

Shocks of this kind (a Fed pivot, a lockdown, a credit event) historically resolve within months, so the model reads the deeply discounted high-β losers as mispriced rather than terminally impaired. With π at this extreme, it commits fully to deep-crisis loser-selection, positioning for the same high-β rebound that cluster 3 harvests during recovery. Cluster 4 is therefore the anticipatory counterpart to cluster 3's realized one, activating at the panic peak on the premise that the panic resolves quickly. When that premise fails, as in the 32-month 2000–2002 dot-com bear, the high-β losers do not rebound and the cluster-4 bet turns into a deep drawdown. Section 5.3.3 traces this reversal-failure mode.

Cluster 4 earns the highest Sharpe in the partition, 1.82, on a mean monthly return of +3.13%. The rebound bet pays off in 13 of the 21 cluster-4 months (62%), with winners comfortably outsizing losers: the high hit rate plus asymmetric magnitudes drive the Sharpe.[20]

5.2.3 Discussion

The four clusters trace a single market cycle. In stages 1 and 2 (continuation), the strategy looks like classical momentum, with above-average winner selection in cluster 1 and a weakened version in cluster 2. In stages 3 and 4 (reversal), it looks nothing like classical momentum, inverting to loser selection in cluster 3 and to a term-structure-wide reversal in cluster 4. The regime-conditioning architecture is therefore not "momentum plus a regime control" but "momentum where momentum works, reversal where momentum crashes".

[20] Cluster-level leg-beta and SHAP-share diagnostics for the full $K = 4$ partition are in Appendix H.7.

XGB ranks stocks on predicted forward return rather than on a per-horizon momentum score, which allows the long leg to hold stocks not top-ranked at any individual horizon. The cluster analysis makes this concrete, with the long leg in clusters 3 and 4 sitting below the cross-sectional mean at every horizon, and in cluster 3 doing so even while the cross-sectional mean itself rallies. No single-horizon rank-based momentum strategy would select these stocks for the long side. The same compositional flexibility implements a regime-conditioned direction flip, classical winner selection (above-average z-scores) in calm and reversal (below-average z-scores) in panic, with the long-short portfolio fully invested throughout.

The analysis makes two contributions to the momentum literature. First, the methodological combination is itself novel. Prior regime-conditional momentum work has paired simple regime treatments (binary up-or-down indicators, single-variable volatility thresholds, or fast-slow blending weights) with linear or rank-based cross-sectional rules. This thesis instead pairs a continuous probabilistic regime filter built from multiple stress indicators (the Bayesian HMM filtered probability π_t^{filter}) with a nonlinear cross-sectional model (XGBoost). The filtered probability acts as a routing variable through tree splits, conditioning the joint use of all twelve momentum horizons on the current market state.

The combination enables a novel class of cross-sectional selection. The selection can act on the full term-structure shape rather than per-horizon rank, and can take a different functional form across regimes. Classical (Jegadeesh and Titman, 1993), multi-horizon (Novy-Marx, 2012), and multi-asset (Asness et al., 2013) momentum strategies cannot produce these compositions by construction, and ML-based asset-pricing methods that select on predicted return (Gu et al., 2020; Beckmeyer and Wiedemann, 2025) do not map their picks against the full term structure of past returns.

Second, the empirical analysis characterizes how the regime-momentum mechanism manifests at a granularity the prior literature does not provide. The fact that cross-sectional momentum profits depend on the prior market state is itself documented descriptively in Cooper et al. (2004). This thesis adds the concrete form: a single nonlinear model learned the routing between continuation- and reversal-mode selection from data alone, on a continuous regime probability rather than a binary backward-looking state. The four-cluster decomposition above quantifies this routing at the z-score level, from roughly $+0.3\,\sigma$ to $+0.8\,\sigma$ above the cross-sectional mean in calm states to uniformly below it across all twelve horizons at panic peaks.

Existing crash-management approaches respond to panic by adjusting how much the strategy invests (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015), how the horizons are blended (Goulding et al., 2023), or by learning regime-agnostic signal reweightings (Beckmeyer and Wiedemann, 2025). XGB instead changes which stocks the strategy invests in. The timing of the response also differs: where Goulding et al. adapt to a more prominent short-horizon signal once trailing returns reveal a cycle change, this thesis identifies panic from real-time stress indicators and positions for the reversal at the onset of stress rather than after the cycle has shown up in prices.

Cluster-level CAPM betas confirm the architectural mechanism. XGB's long leg holds high-β stocks in every cluster, including at panic peaks, so the strategy participates in market rebounds rather than being crushed by them (Appendix H.7, Table 27).

5.3 Robustness and Limitations

The strategy's headline performance comes with three limitations. First, the baseline volatility is high (around 19.5%) and may require down-scaling for risk-averse investors. Section 5.3.1 traces the risk-return trade-off offered by a tunable risk-aversion control. Second, the 2011–2024 test sample does not include the worst historical period for the proposed XGB strategy. Section 5.3.2 extends the evaluation to 25 years (2000–2024) via expanding-window backtests covering the dot-com bust, the GFC, and the 2009 momentum crash. Third, the panic-regime reversal rests on the premise that price declines during stress are temporary. A sustained economic deterioration would expose the strategy to reversal failure, which Section 5.3.3 discusses alongside the complementary D&M exposure-scaling overlay.

5.3.1 Risk-Return Trade-off

The first limitation is volatility itself. An investor may prefer lower portfolio risk at the cost of some expected return. We incorporate a risk-aversion parameter γ into the strategy and trace the out-of-sample trade-off across $\gamma \in [0, 1]$.

The natural starting point is the mean-variance utility $r - \frac{\gamma}{2}\sigma^2$ (Markowitz, 1952), but two issues arise in a cross-sectional ranking setting. First, individual-stock annualized volatilities span roughly 15% to 100% across the universe of stocks, so the σ^2 penalty differs by a factor of about 44 between the low- and high-vol ends. For a moderately risk-averse investor that is a much steeper differential than warranted. A logarithmic penalty $\gamma \log \sigma$ instead gives a constant proportional treatment: a one-percent relative change in σ costs the same whether σ starts at 15% or 100%. Second, raw XGBoost return predictions have a long-tailed distribution, and a single extreme forecast can dominate a utility in which $\hat{r}$ enters linearly. A sign-preserving log transform $\text{sign}(\hat{r}) \log(1+|\hat{r}|/\varepsilon)$ dampens this without changing the ordering of moderate predictions; at $\gamma = 1$ the resulting specification reduces to a log-utility form. We adopt this log-utility-inspired specification for both reasons.

One further change is required for the long-short construction. With a single utility, a rank-based decile split would pile low-volatility names into the long leg and high-volatility names into the short leg as γ rises, raising realized long-short volatility rather than lowering it. Scoring utility separately for each trading direction, with the same volatility penalty applied in both directions, fixes this. A volatile stock receives the same penalty whether held long or sold short. For each stock s, the utility under each trading direction is

$$\begin{aligned} u^s_{\text{long}} &= +\operatorname{sign}(\hat{r}^s)\, \log\bigl(1 + |\hat{r}^s|/\varepsilon\bigr) \;-\; \gamma\, \log(\hat{\sigma}^s), \\ u^s_{\text{short}} &= -\operatorname{sign}(\hat{r}^s)\, \log\bigl(1 + |\hat{r}^s|/\varepsilon\bigr) \;-\; \gamma\, \log(\hat{\sigma}^s), \end{aligned} \tag{16}$$

where $\hat{r}^s$ is the XGBoost forward-return prediction, $\hat{\sigma}^s$ a forward-volatility forecast from a parallel XGBoost model trained on realized 12-month volatility, and $\varepsilon = 0.01$. Each month the model picks stocks that maximize these utilities. Longs are the top decile of u_{long} across the universe of stocks, shorts are the top decile of u_{short}. Full specification, including the volatility-forecast model, is in Appendix F.

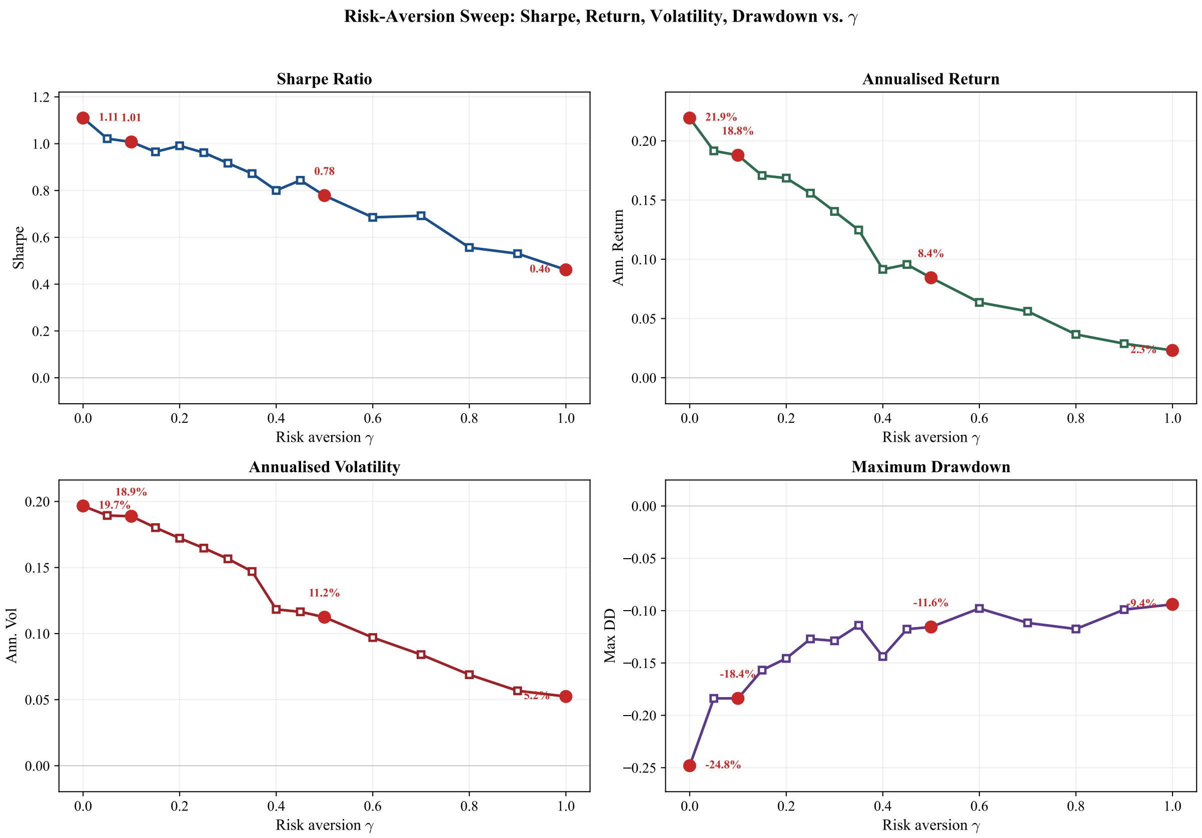

Figure 11: Risk-aversion sweep. Sharpe ratio, annualized return, annualized volatility, and maximum drawdown as a function of γ over $[0, 1]$. Red markers highlight $\gamma \in \{0, 0.1, 0.5, 1.0\}$.

Figure 11 traces the resulting risk-return trade-off as γ rises, where each point corresponds to a different γ and gives a different return-and-risk pair. At $\gamma = 0$ the volatility penalty vanishes and the utilities reduce to a monotonic transformation of the predicted return, so the picks approximately reproduce the headline XGB scoring (Sharpe 1.11; volatility 19.7% and MDD -25% versus 19.5% and -23% in Table 2, the small gap reflecting the per-direction utility transformation). At $\gamma = 0.1$, Sharpe stays at 1.01 while maximum drawdown tightens from −25% to −18%, a near-free reduction in tail risk. At $\gamma = 0.5$, annualized volatility halves from 19.7% to 11.2%, drawdown halves to −12%, and Sharpe settles at 0.78. At $\gamma = 1.0$, volatility falls to 5.2% (roughly a quarter of the market's) and Sharpe remains positive at 0.46. XGB's regime-conditional edge depends on taking volatility. Its reversal bet, especially cluster 4, loads the long leg with stocks that have fallen the furthest. These are typically the most volatile names in the cross-section, and the bet pays off when they rebound. As γ rises, the volatility penalty filters out exactly these names, and the strategy progressively loses the contrarian bet that generates its panic-regime Sharpe. Beyond $\gamma \approx 1.5$ the long-short spread collapses into dollar-neutral noise, comparable to the zero-Sharpe outcome of the linear baselines on the same features (Section 5.1).

5.3.2 Historical Stress: Out-of-Sample Evidence

The 14-year test sample used so far (2011–2024) misses the worst momentum-relevant stress events: the dot-com bust, the global financial crisis, and the 2009 momentum crash documented by Daniel and Moskowitz (2016). To extend the evaluation, the full pipeline (HMM + XGBoost) is re-estimated on an annual expanding-window schedule: annual retraining is initialized in 2000, with each year's model trained only on data available through 31 December of the prior year and used to make predictions for the following year. Stitching these year-by-year predictions together produces a 25-year out-of-sample monthly return series spanning 2000–2024 (299 months).[21] Methodology details are in Appendix H.2.

		XGB			Market		
Period	N months	Sharpe	Total Return	Max DD	Sharpe	Total Return	Max DD
Full sample, 2000–2024	299	0.50	898.4%	−64.8%	0.56	572.6%	−50.3%
2000–2002 dot-com	36	−0.15	−35.9%	−64.8%	−0.72	−38.1%	−45.0%
2003–2006 bull	48	0.70	37.5%	−12.7%	1.70	80.5%	−4.6%
2007–2009 GFC	36	0.26	11.1%	−36.3%	−0.15	−13.9%	−50.3%
2010–2019 calm	120	0.67	155.4%	−17.9%	1.05	258.1%	−17.6%
2020–2024 COVID era	59	1.23	299.1%	−22.2%	0.81	95.2%	−24.8%

Table 8: Sub-period decomposition of the expanding-window out-of-sample backtest, 2000–2024. Annual retraining is initialized in 2000 using all data available through the prior December. Market is the value-weighted CRSP market (Mkt-RF + RF, Fama-French monthly factors).

Worst observed stress: the dot-com bust

The 2000–2002 dot-com bust produced the strategy's largest historical drawdown of −64.8% peak-to-trough. Restricted to the bear-market proper (March 2000–October

[21] The expanding-window construction enforces strict out-of-sample status throughout: each year's predictions are produced by a model trained only on observations available at the corresponding cutoff, so no future information leaks into any prior-year prediction.

2002), the strategy lost −43.8% cumulative over 32 months (Sharpe −0.39). This is the historical realization of the reversal-failure vulnerability discussed in Section 5.3.3 below: a prolonged sector-rotating bear in which beaten-down stocks did not promptly recover, so the panic-regime reversal selection bled.

Stress at the targeted failure mode: the 2009 momentum crash

The single most informative episode is March–December 2009, the period Daniel and Moskowitz (2016) document as a severe momentum crash. Under the same long-short construction used throughout this thesis (NYSE breakpoints, value-weighted, 10 bps cost), unconditional 12-month momentum lost −62.6% cumulative during those ten months. Under expanding-window out-of-sample, the regime-conditional strategy gained +20.5% cumulative (Sharpe 0.73) during the same window. Across the full GFC sub-window (2007–2009 in Table 8) the strategy returned +11.1% (Sharpe 0.26) while the market lost −13.9% (Sharpe −0.15). This is direct historical evidence that the regime mechanism survives the specific stress event that defeats both unconditional momentum and the broader market.

5.3.3 Reversal Failure and the D&M Complement

The dot-com episode above is the empirical realization of the strategy's reversal-failure mode. The cluster 4 bet rests on the premise that panic-period declines are temporary, which holds in shorter panics; XGB earned +29.7% during the COVID crash (February–April 2020) and +20.8% during the 2022 bear market (January–October 2022), and in both cases the decline reversed within months. The premise fails in a sustained economic deterioration where beaten-down stocks do not promptly recover because their businesses are failing, as in the 32-month dot-com bear (March 2000–October 2002) where XGB lost −43.8% cumulative with a peak-to-trough drawdown of −64.8%. The regime classification was correct. What changed was the typical panic-regime pattern. Normally past losers in panic rebound. In the dot-com bust they did not, because the underlying businesses were failing.

XGB and Daniel and Moskowitz (2016) use regime information along complementary channels. XGB conditions *which* stocks the portfolio holds on the regime. D&M condition *how much* the portfolio holds on the regime. On the same data and transaction costs, XGB (Sharpe 1.11, CAPM alpha 23.4%) outperforms D&M managed momentum (Sharpe 0.49, CAPM alpha 10.3%). The two approaches are not substitutes. A hybrid that adds a D&M-style exposure-scaling overlay on top of XGB as a circuit breaker (Section 6.3) would reduce the risk of a total collapse during a prolonged bear market like the dot-com bust without sacrificing XGB's stock-selection advantage in shorter panics.

5.3.4 International Robustness

A natural question for any single-market result is whether the mechanism extends to other developed equity markets, or is specific to the US institutional and structural setting in which it was estimated. We test this by replicating the full pipeline on UK and Japanese equities (Compustat Global, 1990–2024).[22]

	UK				JP			
Strategy	Sharpe	Ann. Ret	Total Ret	Max DD	Sharpe	Ann. Ret	Total Ret	Max DD
Local market	0.52	5.5%	110.5%	−25.2%	0.76	10.7%	313.8%	−21.5%
Fixed 12-mo mom	0.35	6.1%	127.6%	−62.6%	−0.01	−1.9%	−23.0%	−66.3%
XGB	0.72	13.0%	449.2%	−21.4%	0.57	7.0%	155.1%	−19.8%

Table 9: International cross-sectional model performance, 2011–2024 OOS. Regionally-fitted HMM (DD+VOL+REL_N).

The mechanism replicates in both regions. XGB attains positive Sharpes that exceed unconditional 12-month momentum in both markets, and maximum drawdowns compress to similar magnitudes (−22.8% US, −21.4% UK, −19.8% JP) against fixed-momentum drawdowns of −62.6% to −72.2%. The cross-sectional model architecture itself transfers across all three markets without modification.[23]

The Japan result is the most informative of the three. Baseline cross-sectional momentum is essentially absent there (Asness et al., 2013), with unconditional 12-month momentum delivering a Sharpe of −0.01 over 2011–2024. Yet XGB still produces a positive Sharpe of 0.57 by re-ranking the cross-section conditional on the regime. Where the unconditional momentum trade fails, the regime-conditioned version succeeds. This rules out the explanation that XGB is simply a more efficient implementation of an already-profitable signal.

The UK result complements Japan from the opposite end of the baseline-momentum spectrum. Unconditional 12-month momentum is profitable in the UK (Sharpe 0.35), and XGB more than doubles it to 0.72 while cutting the maximum drawdown from −62.6% to −21.4%. The same shape of improvement as in the US, only with a higher baseline.

The answer to the opening question is therefore that the mechanism extends to other developed equity markets. It produces alpha both where unconditional momentum already works (US and UK) and where it does not (Japan), and the panic-side drawdown compression is consistent across all three markets. What is US-specific is the choice of HMM input features that identify panic episodes (the four-feature set DD+DISP+REL_N+CS), not the cross-sectional re-ranking that follows once a regime probability is in hand.

[22] Pipeline unchanged from the US specification except for region-specific HMM features and unconditional decile breakpoints in place of NYSE-equivalent tiers (Section 4, Appendix K).

[23] Only the upstream HMM stress indicators had to be re-selected per region, because US-specific signals such as the Moody's BAA–AAA spread do not have direct equivalents in UK or JP markets.

6 Conclusions

This thesis asked how cross-sectional momentum reorganizes across market regimes. The prior literature established that momentum is regime-dependent and characterized that dependence partially, but not at the level of individual stock selection. The two-stage framework developed here answers the question concretely. A continuous probabilistic regime signal routes a nonlinear cross-sectional model between classical winner selection in calm and term-structure-wide reversal at panic peaks.

6.1 Conclusions and Contributions

Cross-sectional momentum's regime-dependent structure is real and admits a concrete characterization. The regime signal acts as a context variable, conditioning how a nonlinear cross-sectional model uses the full momentum term structure rather than predicting returns directly. The three-method ladder (DET, LR, XGB) introduced in Section 3 establishes that this conditioning is irreducibly nonlinear: the deterministic regime-adaptive formula and the linear classifier deliver marginal or negative Sharpes on the same features that XGB compounds into a Sharpe of 1.11, so nonlinearity is the limiting factor, not the conditioning variable itself. The analysis makes two contributions to the momentum literature, developed in Section 5.2.3.

First, the methodological combination of a continuous probabilistic regime filter with a nonlinear cross-sectional model is novel relative to prior regime-conditional momentum work, which paired simpler regime treatments (binary indicators, threshold-based blends, or fast-slow weights) with linear or rank-based cross-sectional rules. Routing a tunable nonlinear model on a continuous, real-time stress probability is a structurally new approach to the regime-momentum problem, and it enables the empirical characterization developed below.

Second, the empirical analysis characterizes how the cross-section reorganizes across regimes at the level of individual stock-month picks. A four-cluster decomposition traces a market cycle from classical winner selection in calm ($+0.3\,\sigma$ to $+0.8\,\sigma$ above the cross-sectional mean) to term-structure-wide reversal at panic peaks ($-0.7\,\sigma$ to $-0.9\,\sigma$ below across all twelve horizons), a novel composition no single-horizon rank-based strategy can produce. Cooper et al. (2004) noted this state-dependent asymmetry descriptively from a binary backward-looking market state; this thesis implements the regime-conditioned response as a single nonlinear model with the long-short portfolio fully invested throughout, with composition rather than exposure absorbing the regime change.

6.2 Limitations

Reversal-failure vulnerability and test-window scope

The strategy's primary vulnerability is a prolonged economic deterioration in which the prior decline reflects genuine business failure rather than temporary mispricing, so the beaten-down stocks the model selects do not recover. The cluster-4 bet works in short panics but fails in a sustained bear like the 32-month 2000–2002 dot-com bust, where XGB lost −43.8% cumulatively with a −64.8% peak-to-trough drawdown (Section 5.3.2). The regime classification was correct in that episode; what changed was the panic-regime pattern, with businesses failing rather than prices being temporarily depressed, and XGB cannot distinguish these cases ex ante. The 2011–2024 main test window avoids this failure mode, and the expanding-window backtest documents the dot-com failure but offers only one prolonged bear market. Therefore we cannot quantify how the strategy would behave across the distribution of such episodes.

A test window containing a prolonged bear market would deliver a different point estimate. The D&M-style exposure overlay in Section 5.3.3 addresses the structural vulnerability as a circuit breaker. A depression-style scenario lies outside the 25-year historical expanding-window test. This scenario is hard to assess via simulation as an accurate stress test would have to model the heterogeneous business-failure dynamics that drive the failure mode, not just shock all stock returns uniformly by a chosen magnitude over a sustained period.

Frozen HMM parameters and regime drift

The HMM is estimated on 1990–2010 and held fixed during the test period. This assumes that the statistical properties of calm and panic regimes remain stationary. If the market undergoes structural change (a permanent increase in baseline volatility due to algorithmic trading or shifting monetary policy regimes, for example), the trained thresholds may become miscalibrated, with structurally calm periods classified as panic. The sustained elevation of π_t^{filter} in recent years may partly reflect such drift rather than genuine prolonged stress. Online re-estimation, in which the HMM updates as new data arrives, would address this directly.

Specification search

The HMM feature set (DD+DISP+REL_N+CS) was selected by comparing candidates on out-of-sample portfolio Sharpe in Pass 2 of the four-pass procedure described in Section 4.1. Because feature selection and headline performance share the same out-of-sample window, the reported Sharpe of the selected combination is subject to data-snooping bias. Table 11 bounds the size of that bias on two fronts. First, the top candidate combinations cluster within a narrow Sharpe range, with several four-feature alternatives close to the selected one. Second, when the same feature combinations are re-evaluated across random subsets of the seed pool, the selected combination remains the highest in the table, so its lead is not an artefact of which seeds happened to be drawn in Pass 2. The chosen feature set is therefore one of several near-equivalent specifications, and the data-snooping bias is bounded above by the small Sharpe gap between the chosen combination and its closest alternatives in the table.

6.3 Recommendations for Future Research

Five directions extend this framework most directly.

Richer regime-signal representation

The current model uses only the contemporaneous filtered probability π_t^{filter}, discarding regime dynamics. Two complementary extensions follow. The first looks forward, computing a k-step regime forecast $P(s_{t+k} = 1 \mid \mathbf{z}_{1:t})$ from the HMM's transition probabilities so the portfolio can reposition before a regime shift rather than after it. The gain is concentrated in transition months, where the current framework trades on a stale view of the regime. The second looks backward, feeding the trajectory of π_t^{filter} rather than its current value so that a fresh spike (where mean reversion is most likely) and a sustained elevation (where the reversal mechanism may be breaking down) become distinguishable. A sequential model, or simple lag-based feature engineering (lagged π_t^{filter} values, a short-window rate of change, a regime-duration counter), would expose this trajectory. Together, the two extensions provide anticipation and memory.

Alternative nonlinear models

The selection rule found here is what XGBoost can learn from the regime probability and the twelve momentum horizons; a more expressive model may recover a sharper version of the same signal. The most natural alternative is a feedforward neural network, which Gu et al. (2020) find competitive with tree ensembles for cross-sectional return prediction, with performance peaking at three hidden layers. The primary goal is signal optimization: a network trained on the same inputs could capture interactions that a depth-4 tree cannot represent, raising headline performance and producing sharper cluster boundaries. A secondary benefit is robustness: if the regime-conditional shift recurs in a non-tree architecture, the mechanism is a property of the data rather than an artefact of XGBoost's tree-split structure, and comparing the network's per-feature attribution to the SHAP decomposition in Table 6 would reveal whether the regime signal plays the same routing role outside the tree representation.

Short-leg structure

The empirical characterization in Section 5 is built on long-leg cross-sectional z-curves; the short leg enters only by symmetric decile construction and is not characterized in its own right. Appendix H.7 reports that the short basket is structurally distinct from a negated long leg, with short-vs-negated-long correlation collapsing at panic peaks (cluster 4). This indicates that the short leg follows different selection modes than the long leg, with Nagel (2012)'s short-horizon reversal mechanism a plausible driver of the cluster-4 divergence. Re-running the cluster decomposition directly on the short-leg z-curves, rather than relabeling long-leg clusters, would recover any short-side modes the current characterization misses. The analysis was deferred as beyond the scope of this thesis, since the parallel decomposition would not have changed the headline mechanism documented from the long leg.

Regime-conditional feature sets

Section 5.1.3 shows that adding firm-level fundamentals to XGB hurts in panic while leaving calm performance roughly unchanged, suggesting the optimal feature set may itself be regime-dependent. A layered construction in which XGB produces the initial long-short ranking and fundamentals enter as a post-ranking filter (excluding bottom-quintile-quality names from the long leg, high-leverage names from the short leg) is a natural starting point. A more ambitious variant trains separate cross-sectional models for the calm and panic branches, each with its own feature set. Both designs require careful out-of-sample validation because the regime-conditional split reduces effective sample size.

Exposure-scaling overlay (D&M complement)

A D&M-style exposure-scaling overlay layered on top of XGB's stock selection would provide a circuit breaker against sustained downturns that violate the panic-rebound premise the model assumes. The overlay requires a triggering rule; candidates include π_t^{filter} above a threshold for a minimum number of consecutive months, or its rolling maximum over a fixed window exceeding a calibrated bound. Preliminary checks on the 2011–2024 sample show no improvement, because the sample contains no prolonged bear market and the overlay only reduces exposure during months that turn out to be profitable. Its value would emerge in a sustained downturn outside the current window; the dot-com sub-period of the 25-year expanding-window backtest (Section 5.3.2) is the appropriate setting to evaluate candidate overlay rules.

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Appendix A: Data Sources and Variable Definitions

Three primary data sources are used. **CRSP** (Center for Research in Security Prices, via WRDS) provides monthly stock-level returns, prices, shares outstanding, exchange codes, and share codes from the Monthly Stock File (`crsp.msf`), as well as daily market returns from the Daily Stock Index file (`crsp.dsi`). **Compustat** (via WRDS) provides quarterly fundamental accounting data from `comp.fundq`, linked to CRSP via the CCM bridge table using primary link types (LU, LC). **FRED** (Federal Reserve Economic Data) provides Moody's BAA and AAA corporate bond yields for computing the credit spread.

Following Shumway (1997), delisting returns are incorporated to prevent survivorship bias. When a stock is delisted, its actual delisting return is used if available; for performance-related delistings (CRSP codes 500–584) with missing returns, −30% is imputed.

Table 10 summarises all data variables.

Variable	Source	Description	Used in
Stock-level variables			
Monthly returns	CRSP `msf`	Adjusted holding-period returns	Sections 3, 5
Price, shares out.	CRSP `msf`	For market equity computation	Section 4
Exchange, share codes	CRSP `msf`	Universe filtering (codes 10, 11)	Section 4
Delisting returns	CRSP `msedelist`	Survivorship bias correction	Section 4
Book equity (`ceqq`)	Compustat `fundq`	Book-to-market, ROE	Appendix L
Total assets (`atq`)	Compustat `fundq`	Asset growth, gross profit, CF ratios	Appendix L
Income (`ibq`)	Compustat `fundq`	ROE, earnings growth, accruals	Appendix L
Revenue (`revtq`)	Compustat `fundq`	Gross profitability	Appendix L
COGS (`cogsq`)	Compustat `fundq`	Gross profitability	Appendix L
Debt (`dlttq`, `dlcq`)	Compustat `fundq`	Leverage ratio	Appendix L
Operating cash flow (`oancfy`)	Compustat `fundq`	CFO, FCF, accruals	Appendix L
Capital expenditure (`capxy`)	Compustat `fundq`	Free cash flow	Appendix L
Market-level variables (HMM inputs)			
Daily market returns	CRSP `dsi`	Drawdown (DD) computation	Section 4.2
Market price index	CRSP `dsi`	Drawdown (DD)	Section 4.2
Cross-sect. stock returns	CRSP `msf`	Dispersion (DISP), REL_N	Section 4.2
BAA, AAA bond yields	FRED	Credit spread (CS)	Section 4.2
Derived features			
$mom_1, \ldots, mom_{12}$	Computed	12 trailing momentum signals	Section 4.4
π_t^{filter}	HMM output	Filtered panic probability	Section 4.3
10 fundamentals (ablation only)	Computed	bm, roe, gross profit, cash flow, etc.	Appendix L

Table 10: Summary of all data variables used in the analysis.

Notes: All data accessed via WRDS. Compustat linked to CRSP via the CCM bridge table using primary link types (LU, LC). Fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. The sample covers January 1990 to November 2024 (training: 1990–2010, 240 months; test: 2011–2024, 167 months).

Appendix B: HMM Feature Selection Tables

The two tables in this section support the four-pass feature selection procedure described in Section 4.1. Table 11 reports the top 10 candidate feature combinations on ensemble Sharpe and cross-seed stability. Table 12 reports include-one and leave-one-out ablations of the production four-feature set.

D	Combination	Sharpe	Mean	σ
4	DD+DISP+REL_N+CS	0.91	0.84	0.11
4	DD+DISP+CS+TERM	0.82	0.55	0.15
4	DD+DISP+REL_N+SKEW	0.80	0.69	0.08
4	DD+VOL+DISP+REL_N	0.80	0.77	0.09
3	DD+DISP+REL_N	0.77	0.77	0.10
4	DD+VOL+DISP+TERM	0.71	0.67	0.16
3	DD+VOL+CS	0.69	0.43	0.13
1	DD	0.68	0.67	0.14
4	DD+DISP+REL_N+TERM	0.62	0.45	0.18
3	DD+ADR+TERM	0.57	0.69	0.07

Table 11: Top 10 feature combinations from the four-pass selection pipeline (long-short, 10 bps costs). D = number of HMM features. Sharpe = ensemble Sharpe ratio (10 HMM seeds, 20 XGBoost seeds). Mean and σ are the mean and standard deviation of the Sharpe ratio across 20 independent experiments with different seed partitions, measuring robustness to random initialisation. DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean.

HMM Input Features	XGB Sharpe	Δ vs Full
Full 4F baseline (DD+DISP+REL_N+CS)	0.97	—
Include-one (single-feature HMM)		
DD only (market drawdown)	0.66	−0.31
Leave-one-out (three-feature HMM)		
Drop DD	0.51	−0.46
Drop DISP	0.96	−0.01
Drop REL_N	0.43	−0.54
Drop CS	0.96	−0.01

Table 12: HMM input feature ablation (long-short, feature-selection seed counts). The absolute Sharpe levels differ from the production model (Table 2) due to fewer seeds; the relative comparisons (Δ) are the relevant quantities. DD alone is the strongest single-feature classifier but trails the full set by −0.31. Removing DD or REL_N causes the largest leave-one-out drops (−0.46 and −0.54), while DISP and CS contribute marginally to the Sharpe but improve seed stability (Section 4.1).

Appendix C: Portfolio Formation, Transaction Costs, and Performance Metrics

C.1 Portfolio Construction

For each method, a long-short portfolio is constructed each month. Stocks are ranked by their scores, and NYSE-listed stocks are used to determine decile breakpoints. All stocks (including AMEX and NASDAQ) whose score exceeds the 90th percentile breakpoint enter the long leg; all stocks below the 10th percentile breakpoint enter the short leg. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum's cross-sectional stock selection from market beta, providing a cleaner test of the strategy's ability to distinguish future winners from losers.

Portfolio weights within each leg are assigned proportionally to market equity (value-weighting):

$$w_t^{s,L} = \frac{ME_t^s}{\sum_{s \in \mathcal{L}_t} ME_t^s}, \qquad w_t^{s,S} = \frac{ME_t^s}{\sum_{s \in \mathcal{S}_t} ME_t^s}, \tag{17}$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of stocks in the long and short legs at month t, respectively. The long-short portfolio return is:

$$r_t^{\mathrm{LS}} = \sum_{s \in \mathcal{L}_t} w_t^{s,L} \cdot r_t^s - \sum_{s \in \mathcal{S}_t} w_t^{s,S} \cdot r_t^s. \tag{18}$$

Value-weighting tilts each leg toward large, liquid stocks, reducing the influence of small-cap return anomalies and improving implementability.

C.2 Transaction Costs

All portfolios are rebalanced monthly.

A one-way transaction cost of $c = 10$ basis points is applied to the turnover in both the long and short legs at each rebalancing date. The net portfolio return at month t is:

$$r_t^{\mathrm{net}} = r_t^{\mathrm{gross}} - c \cdot (\tau_t^L + \tau_t^S), \tag{19}$$

where τ_t^L and τ_t^S are the one-way turnover in the long and short legs, respectively:

$$\tau_t^L = \frac{1}{2} \sum_s \left| w_t^{s,L,\mathrm{new}} - w_{t-1}^{s,L,\mathrm{prev}} \right|, \qquad \tau_t^S = \frac{1}{2} \sum_s \left| w_t^{s,S,\mathrm{new}} - w_{t-1}^{s,S,\mathrm{prev}} \right|, \tag{20}$$

and each summation runs over all stocks appearing in either the new or previous portfolio for that leg. The 10 basis point cost reflects the transaction environment for a value-weighted, large-cap-biased strategy where NYSE breakpoints tilt toward liquid securities.

Transaction costs are applied *post-hoc* as an accounting adjustment and are not incorporated into the models' training objectives. The cross-sectional models (DET, LR, and XGB) optimize stock-level prediction (classification accuracy or return forecasting) without knowledge of the previous portfolio, the resulting turnover, or the cost of rebalancing. This is standard practice in the empirical asset pricing literature (e.g., Fama and French, 2015; Goulding et al., 2023), where portfolio construction is treated as a fixed downstream step applied uniformly to all strategies. An end-to-end framework that jointly optimizes stock selection and portfolio-level objectives (e.g., net-of-cost Sharpe ratio) would require a fundamentally different architecture, such as a reinforcement learning or differentiable portfolio optimization layer, and remains a potential extension of this work.

C.3 Performance Metrics

Standard Metrics

Strategy performance is assessed using four standard risk-adjusted metrics computed on monthly net-of-fee returns.

Annualized return The geometric annualized return over n months is:

$$R_{\text{ann}} = \left[\prod_{t=1}^{n}(1 + r_t)\right]^{12/n} - 1. \tag{21}$$

Annualized volatility The annualized standard deviation of monthly returns:

$$\sigma_{\text{ann}} = \sigma(r) \cdot \sqrt{12}. \tag{22}$$

Sharpe ratio The annualized Sharpe ratio:[24]

$$SR = \frac{\bar{r}}{\sigma(r)} \cdot \sqrt{12}, \tag{23}$$

where $\bar{r}$ is the mean monthly return. Sharpe is computed from the arithmetic monthly mean annualized by $\sqrt{12}$, while annualized returns reported elsewhere in this thesis are compound (geometric); under high volatility the two definitions of average return differ, so Sharpe should not be inferred from a table by dividing annualized return by annualized volatility (e.g. at depth 1 in the methodology hyperparameter sweep, the reported Sharpe is 0.53 but $8.7\%/19.4\% = 0.45$).

Maximum drawdown Let $C_t = \prod_{\tau=1}^{t}(1 + r_\tau)$ denote the cumulative wealth path.

[24] All Sharpe ratios use raw returns. The risk-free rate averaged ∼2% over 2011–2024 (near zero pre-2021), so absolute Sharpe ratios are overstated by roughly 0.05–0.10; relative comparisons are unaffected because the convention is uniform.

The maximum drawdown is:

$$MDD = \min_{1 \le t \le n} \frac{C_t - \max_{1 \le \tau \le t} C_\tau}{\max_{1 \le \tau \le t} C_\tau}. \tag{24}$$

Regime-Conditional Evaluation

To assess whether the regime signal adds value beyond unconditional stock selection, performance is decomposed by market regime. Months are partitioned into calm ($\pi_t^{\text{filter}} < 0.5$) and panic ($\pi_t^{\text{filter}} \ge 0.5$) periods, and the Sharpe ratio is computed separately within each subset:

$$SR_{\text{calm}} = \frac{\bar{r}_{\text{calm}}}{\sigma(r_{\text{calm}})} \cdot \sqrt{12}, \qquad SR_{\text{panic}} = \frac{\bar{r}_{\text{panic}}}{\sigma(r_{\text{panic}})} \cdot \sqrt{12}. \tag{25}$$

This decomposition helps assess whether the strategy's alpha originates primarily from calm-period stock selection, panic-period risk avoidance, or both.

Per-Cluster Sharpe

The cluster Sharpes reported in Section 5.2.2 apply the standard Sharpe formula to the months in each cluster. Let M_C denote the set of months assigned to cluster C, let r_t be the long-short monthly portfolio return at month t, and let $n_C = |M_C|$. The point-estimate cluster Sharpe is

$$\text{Sharpe}_C = \frac{\bar{r}_{M_C}}{\sigma(r_{M_C})} \cdot \sqrt{12},$$

where

$$\bar{r}_{M_C} = \frac{1}{n_C} \sum_{t \in M_C} r_t, \qquad \sigma(r_{M_C}) = \sqrt{\frac{1}{n_C - 1} \sum_{t \in M_C} (r_t - \bar{r}_{M_C})^2}.$$

The denominator is therefore the unbiased sample standard deviation of the in-cluster long-short monthly returns; annualisation by $\sqrt{12}$ follows the monthly-to-annual convention used throughout this thesis. Confidence intervals come from a block bootstrap (block size 6, 5,000 reps) on the in-cluster return series.

Appendix D: Feature Importance Methods

D.1 SHAP Values for XGBoost

To interpret the XGBoost model and assess the marginal contribution of each feature, SHAP values (Lundberg and Lee, 2017) are computed on the test set. The SHAP value of feature j for observation $\mathbf{x}$ measures its average marginal contribution to the prediction across all possible feature orderings:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x}) \Big], \tag{26}$$

where N is the full feature set, $p = |N|$, and $f_S(\mathbf{x})$ is the expected model output when only features in S are observed. For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes these values exactly in polynomial time. The overall importance of each feature is summarized as:

$$I_j = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j(\mathbf{x}_i)|. \tag{27}$$

Two aggregations of I_j appear in the main text. The first is the *total |SHAP| share*, each feature's contribution as a fraction of overall importance:

$$\text{share}_j = \frac{I_j}{\sum_{k=1}^{13} I_k}. \tag{28}$$

The 46% figure reported for π_t^{filter} in Section 5.2.1 is share_π computed across the full test set.

The second is the *per-horizon momentum share by leg* plotted in Figure 5, which restricts attention to the twelve momentum horizons within a single leg-regime cell. For leg $\ell \in \{\text{long}, \text{short}\}$, regime $r \in \{\text{calm}, \text{panic}\}$, and horizon $h \in \{1, \ldots, 12\}$,

$$s_{\ell,r,h} = \frac{\sum_{i \in \mathcal{I}_{\ell,r}} |\phi_h(\mathbf{x}_i)|}{\sum_{h'=1}^{12} \sum_{i \in \mathcal{I}_{\ell,r}} |\phi_{h'}(\mathbf{x}_i)|}, \tag{29}$$

where $\mathcal{I}_{\ell,r}$ is the set of test observations in leg ℓ and regime r. By construction the twelve shares within each (ℓ, r) sum to one. Pooling across regimes weighted by observation count gives the leg-level shares displayed in Figure 5:

$$s_{\ell,h} = \frac{\sum_r n_{\ell,r}\, s_{\ell,r,h}}{\sum_r n_{\ell,r}}, \qquad n_{\ell,r} = |\mathcal{I}_{\ell,r}|. \tag{30}$$

Two additional analyses are performed. First, the mean absolute SHAP value of π_t^{filter} is computed separately for calm and panic months to assess whether the regime signal's influence varies across market states. Second, regime-specific XGBoost models are trained on calm-only and panic-only subsets of the training data, and their SHAP feature importances are compared to identify which features the model relies on differentially across regimes.

Appendix E: Alternative Training Targets for XGBoost

Section 5.3.1 argues that embedding a risk penalty directly in the XGBoost training target degrades performance, motivating the per-direction utility specification used in the main text. Table 13 shows this degradation is not an artefact of the mean-variance functional form: the same pattern appears under three families of training targets with varying risk tolerance.

Target family	Parameter	Sharpe	Ann. Ret	Ann. Vol	MDD
Baseline (r)	–	1.107	30.0%	27.1%	−28.3%
Mean-variance: $r - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.061	22.9%	21.6%	−22.3%
	$\gamma = 1$	1.024	16.5%	16.2%	−22.8%
	$\gamma = 2$	1.135	15.5%	13.6%	−22.0%
	$\gamma = 5$	1.062	13.5%	12.7%	−22.5%
	$\gamma = 10$	1.082	13.1%	12.1%	−18.9%
Log mean-variance: $\log(1{+}r) - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.051	14.6%	13.9%	−17.9%
	$\gamma = 1$	1.039	13.8%	13.4%	−23.8%
	$\gamma = 2$	1.080	14.0%	12.9%	−20.9%
	$\gamma = 5$	1.127	14.1%	12.4%	−18.7%
	$\gamma = 10$	0.995	12.1%	12.3%	−20.2%
Sharpe-like: $r\,/\,\sigma^a$	$a = 0.5$	1.148	24.8%	21.4%	−22.4%
	$a = 1.0$	0.984	14.6%	15.0%	−29.1%
	$a = 1.5$	0.936	12.4%	13.5%	−21.0%
	$a = 2.0$	0.759	12.8%	18.1%	−24.5%

Table 13: Out-of-sample performance of XGBoost (XGB) under alternative training targets. The construction here is long-only (top-decile, NYSE-breakpoint, value-weighted), used as a diagnostic that isolates the effect of the training target itself from the long-short construction; the absolute Sharpes are therefore not comparable with the long-short headline in Table 2, only the relative ordering across rows. Every risk-adjusted target reduces returns faster than volatility.

The pattern is consistent: returns fall faster than volatility under every risk penalty, regardless of functional form. The mildest adjustments (mean-variance with $\gamma = 0.5$, Sharpe-like with $a = 0.5$) come closest to the baseline but still underperform. This confirms that the diagnostic in Section 5.3.1 is not specific to the mean-variance utility formulation; the single-model training-target approach is structurally unsuited to incorporating risk aversion in a long-short setting, which the long-and-short utility construction resolves.

Appendix F: Long-Short Utility Construction

This appendix supplements the long-and-short utility specification in Section 5.3.1 with the volatility-forecast model, the high-γ collapse mechanism, and the sweep grid.

Volatility forecast

The expected-return forecast $\hat{r}_t^s$ comes from the XGBoost ensemble described in Section 3. The volatility forecast $\hat{\sigma}_t^s$ comes from a second XGBoost ensemble trained on the realized

12-month forward volatility

$$\sigma^s_{[t+1,\,t+12]} \;=\; \mathrm{std}\big(r^s_{t+1},\ldots,r^s_{t+12}\big),$$

using the same feature set augmented with trailing 12-month realized volatility. The 50-seed ensemble is trained on 1990–2010 and evaluated out-of-sample on 2011–2024.

Convergence to dollar-neutral noise at high γ

Stocks appearing in both top deciles of u_{long} and u_{short} (the overlap set, which grows with γ) are retained on both sides, and their contributions to the long and short books net out in $r_L - r_S$. This is the mechanism by which the long-short spread converges to zero as γ rises beyond the useful range. At high γ, the volatility penalty dominates the return signal, low-$\hat{\sigma}$ stocks occupy both top deciles simultaneously, and the two legs become near-identical value-weighted low-volatility portfolios.

Sweep grid

The risk-aversion sweep in Figure 11 uses a 16-point grid over $[0,1]$: step 0.05 up to 0.5, step 0.1 thereafter.

Appendix G: Robustness Checks

This section reports six robustness checks that complement the main results in Section 5.

G.1 Sub-Period Stability

To assess whether the main findings are driven by a single favorable sub-period, the test sample is split into three roughly equal windows: 2011–2015, 2016–2020, and 2021–2024.

	2011–2015	2016–2020	2021–2024	Full
Market	0.72	1.04	0.73	0.84
Fixed 12-mo	0.74	-0.20	-0.11	0.06
LR	0.47	-0.02	-0.38	-0.01
XGB	0.59	1.04	1.70	1.11

Table 14: Sub-period Sharpe ratios (long-short). The test period is split into three roughly equal sub-periods. XGB is the only strategy with positive Sharpe ratios in all three sub-periods.

In long-short construction, XGB is the only strategy that produces positive Sharpe ratios in every sub-period (0.59, 1.04, 1.70). LR collapses in 2016–2020 (Sharpe -0.02) and turns negative in 2021–2024 (-0.38). Fixed 12-month momentum turns negative after 2015, consistent with the momentum crash mechanism documented by Daniel and

Moskowitz (2016). XGB's strongest performance (1.70) is in 2021–2024, a period characterized by elevated volatility and multiple stress episodes, precisely the conditions where regime-conditional stock selection adds the most value.

G.2 Turnover and Transaction Cost Sensitivity

A strategy's net-of-cost performance depends critically on its turnover. Table 15 reports average monthly one-way turnover for each strategy.

Strategy	Avg. Monthly TO	Annualised TO
Fixed 12-mo	70.1%	842%
Fixed 1-mo	184.3%	2,211%
LR	168.8%	2,026%
XGB	132.3%	1,588%

Table 15: Portfolio turnover by strategy (long-short). Average monthly one-way turnover and annualised turnover (monthly × 12).

In long-short construction, XGB's average monthly one-way turnover is 132.3%, reflecting the regime-conditional changes in stock selection that shift the portfolio composition substantially. Table 16 translates these turnover differences into Sharpe ratio sensitivity across a range of transaction cost assumptions.

	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps
Fixed 12-mo	0.10	0.08	0.06	0.03	0.00	-0.06
Fixed 1-mo	0.39	0.32	0.26	0.14	0.01	-0.24
LR	0.12	0.05	-0.02	-0.16	-0.29	-0.57
XGB	1.19	1.15	1.11	1.03	0.95	0.78

Table 16: XGB remains profitable at 50 bps. All other strategies become unprofitable at moderate cost levels.

XGB remains profitable across all cost levels tested. Even at 50 bps one-way (five times the baseline), its Sharpe ratio is 0.78. All other strategies become unprofitable at moderate cost levels. The baseline 10 bps assumption is reasonable for a value-weighted strategy using NYSE breakpoints, which tilts toward the most liquid names.

G.3 Model Sensitivity

The XGBoost hyperparameters are set on the training period and stress-tested out of sample; tree depth in particular is analysed in Appendix G.4. To verify that the results are not an artefact of a particular configuration, Table 17 reports out-of-sample performance under alternative tree depths, learning rates, and ensemble sizes.

Configuration	Sharpe	Ann. Ret	Ann. Vol
depth=1, lr=0.05, n=500	0.53	8.7%	19.4%
depth=2, lr=0.05, n=500	0.86	15.7%	19.1%
depth=3, lr=0.05, n=500	0.98	19.2%	19.9%
depth=4, lr=0.05, n=500	1.11	21.7%	19.5%
depth=5, lr=0.05, n=500	0.92	17.4%	19.4%
depth=6, lr=0.05, n=500	0.90	17.0%	19.5%
depth=4, lr=0.01, n=500	0.99	18.7%	19.2%
depth=4, lr=0.10, n=500	1.00	19.6%	19.9%
depth=4, lr=0.05, n=200	1.06	20.5%	19.4%
depth=4, lr=0.05, n=1000	0.96	18.7%	19.9%

Table 17: XGBoost hyperparameter sensitivity (long-short). Each row varies one hyperparameter from the baseline. The baseline is not the single best configuration, but all variants outperform unconditional momentum.

Hyperparameter sensitivity

At the baseline depth of 4, the Sharpe ratio ranges from 0.89 to 1.11 across learning rate and ensemble size variations. Performance peaks at depth 4 and falls off on either side (see Appendix G.4 for the depth analysis). The results confirm that XGB's performance is not an artefact of a particular hyperparameter configuration.

G.4 Tree Depth and Interaction Order

The maximum tree depth is the order of feature interactions the model can represent: a decision path of length d conditions on d features jointly. Depth therefore controls whether the regime signal interacts with single momentum horizons or with the joint configuration of several. Depth 4 is the smallest order at which one path can gate on the regime (depth 1) and then combine three momentum horizons (depths 2 to 4), a regime-conditioned read of term-structure shape rather than of one horizon in isolation. This is the interaction the mechanism of Section 5.2.2 turns on.

Figure 12 reports two views of the depth response, both at the production hyperparameters (learning rate 0.05, 500 trees). The left panel is a five-fold expanding-window cross-validation on the training period (1990–2010), each fold trained on an initial window and validated on the following two years, scored by net-of-fee long-short validation Sharpe. The right panel is the out-of-sample sensitivity sweep on the 2011–2024 test window (Table 17). In cross-validation the depth response is flat: validation Sharpe sits between 0.44 and 0.52 across all six depths, a spread far smaller than the across-fold standard deviation of roughly 0.7, so no depth is statistically separable from another in sample. Allowing each depth its own best learning rate and tree count does not change this: the best per-depth configurations still fall within fold noise of one another, and the single best configuration in the entire grid lies at depth 3 (validation Sharpe 0.60 against

0.58 at depth 4), a lead of 0.02 that is negligible against the same noise. Cross-validation cannot resolve the depth.

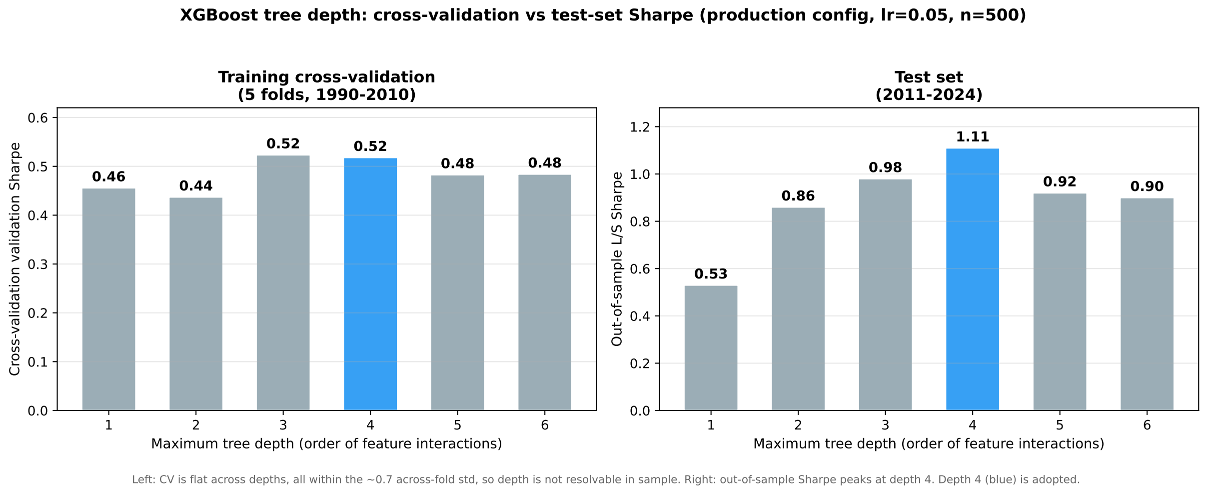

Figure 12: XGBoost tree-depth response at the production hyperparameters (learning rate 0.05, 500 trees). Left: five-fold expanding-window cross-validation validation Sharpe on the training period (1990–2010); the response is flat across depths, all within the across-fold standard deviation of roughly 0.7, so no depth is separable in sample. Right: out-of-sample long-short Sharpe on the 2011–2024 test window, which peaks at depth 4. Depth 4 (blue) is the adopted depth.

On the evaluation window the depth response is more pronounced, though this is the same window on which the headline result is reported: out-of-sample Sharpe rises from 0.53 at depth 1 to a peak of 1.11 at depth 4, then falls to 0.92 and 0.90 at depths 5 and 6. This rise is consistent with the interaction-order account above, the third within-regime level adding signal, while the decline beyond depth 4 is consistent with the added order capturing noise rather than signal. We adopt depth 4 because it is the smallest interaction order that completes the regime-conditioned term-structure read; the out-of-sample peak corroborates this a-priori choice but, coming from the evaluation window, is treated as corroboration rather than as the selection criterion. The mechanism itself is qualitatively the same across depths 3 and 4, so the choice within that pair does not affect the reported results.

Finally, the depth choice is independent of the number of selection modes recovered in Section 5.2.2. The value $K = 4$ is a K-means partition of the long-leg cross-sectional z-curves, computed on the model's monthly picks and chosen on economic grounds; it is not a property of the tree depth, and the same partition could be applied to the picks of a shallower model. The evidence for the regime-momentum mechanism, namely the SHAP attribution, the z-curves, and the leg-level betas, likewise does not rest on the depth. Depth 4 sets the model's capacity to express the mechanism, not the number of modes the downstream clustering reports.

G.5 Number of Hidden States

The baseline HMM uses $K = 2$ states (calm and panic). A natural question is whether finer regime granularity (for example, distinguishing mild corrections from severe crises) would improve the downstream portfolio models. Table 18 reports out-of-sample Sharpe ratios for $K = 2, 3, 4, 5$, each averaged over five representative Gibbs sampler seeds (the production regime signal averages 200 seeds, but five suffice for this diagnostic).

	LR		XGB	
K	Mean Sharpe	Std	Mean Sharpe	Std
2	-0.012	0.000	1.107	0.000
3	-0.029	0.004	0.831	0.164
4	-0.029	0.003	0.606	0.291
5	-0.028	0.003	0.724	0.087

Table 18: Out-of-sample Sharpe ratios for HMMs with $K = 2, 3, 4, 5$ states (long-short). The K=2 row uses the 200-seed production regime signal as the reference point (its Std is zero by construction). For $K = 3, 4, 5$ each entry reports the mean and standard deviation across 5 independent Gibbs sampler seeds (2,000 iterations each). LR is invariant to K. XGB shows no consistent improvement beyond $K = 2$, and cross-seed variability increases with K.

In long-short construction, XGB shows no consistent improvement beyond $K = 2$. Cross-seed variability increases with K, and the intermediate states that emerge for $K \geq 3$ lack the persistence needed to provide a stable conditioning signal. With only 240 training months, the data cannot reliably separate more than two regimes for the purpose of downstream cross-sectional stock selection.

Beyond estimation, $K = 2$ has an interpretability advantage. The two-state model produces a single continuous signal (π_t^{filter}) that XGBoost can condition on in a straightforward way, with higher values indicating more stress. With $K \geq 3$, the additional states may capture dimensions of the economy that are orthogonal to the continuation-reversal mechanism, for example a "growth" state characterized by strong fundamentals but moderate stress indicators. Such a state would conflate two economically distinct situations (a genuine expansion vs. a recovery from panic) and could introduce noise into the regime signal, degrading the model's ability to distinguish continuation from reversal. The two-state framework avoids this by restricting the regime classification to a single dimension, the degree of market stress, which is the dimension most relevant to momentum's cross-sectional structure.

G.6 January Exclusion

Momentum is known to reverse in January (Jegadeesh and Titman, 1993): past losers outperform past winners, partially offsetting momentum's annual profitability. To verify that XGB's performance is not driven by a January seasonal, Table 19 recomputes all metrics after excluding the 14 January months from the test period. XGB's Sharpe ratio declines marginally from 1.11 to 1.06, confirming that the regime-momentum interaction operates throughout the calendar year. Fixed 12-month momentum improves slightly when January is excluded (0.06 to 0.14), consistent with the well-documented January reversal in unconditional momentum.

	Ann. Ret	Ann. Vol	Sharpe	Months
Full sample (baseline)				
XGB	21.7%	19.5%	1.11	167
Fixed 12-mo mom	−1.9%	26.1%	0.06	167
Excluding January				
XGB	20.9%	19.9%	1.06	153
Fixed 12-mo mom	0.1%	25.6%	0.14	153

Table 19: January exclusion test. Performance is recomputed after dropping all 14 January months from the test period.

Appendix H: External Validity

H.1 Baseline Train/Test Split

Table 20 reports the baseline train/test split performance. Robustness across training windows and prediction horizons is provided separately by the 25-year expanding-window backtest (Appendix H.2 below, also reported in Section 5.3.2), which re-estimates the full pipeline annually and therefore subsumes a continuum of alternative splits.

Train / Test split	N_{test}	LR Sharpe	XGB Sharpe
1990–2010 / 2011–2024 (baseline)	167	−0.01	1.11

Table 20: Out-of-sample performance under the baseline train/test split (long-short, mom+π, 50-seed ensemble). LR collapses while XGB achieves a Sharpe of 1.11 with highly significant factor model alphas.

In the baseline split (1990–2010 training, 2011–2024 test), XGB achieves a Sharpe of 1.11 while LR collapses (−0.01), confirming the nonlinear regime-momentum interaction documented in Section 5.1.

H.2 25-Year Expanding-Window Out-of-Sample Methodology

The historical evidence reported in Section 5.3.2 comes from a separate analysis from the one above: a 25-year out-of-sample backtest in which the full pipeline (HMM + XGBoost) is re-estimated every January, with annual retraining initialized in 2000 and each year's predictions made by a model trained only on data available through the prior December. Each retraining uses the production seed configuration described in Section 3 (200 HMM seeds, 50 XGBoost seeds) and the same long-short construction described in Appendix C.1.

The HMM's panic-state label is fixed by an economically motivated sign convention rather than the data-driven percentile comparison used elsewhere. The convention encodes the known direction of each feature in panic regimes: -1 for drawdown, $+1$ for cross-sectional dispersion, -1 for relative market participation, and $+1$ for the credit spread. The data-driven label-switching procedure used in the main pipeline is unstable on short training windows with few observed crisis months (under 20 months in the crisis windows used by the percentile comparison; see Appendix I), so the fixed-sign fallback activates for retrainings before 2002. From 2002 onward the data-driven procedure resumes, by which point the training window contains the dot-com bust as a calibration anchor. The HMM seed loop is parallelised across CPU cores via `multiprocessing.Pool`; the parallel implementation is bit-identical to the serial reference (verified in the test suite by comparing per-seed pi-filter arrays).

Each retraining produces 12 monthly out-of-sample long-short returns, which are stitched into a single monthly return series; this thesis reports the 25-year sub-sample from January 2000 to November 2024 (299 months). Every prediction is strictly causal: the model used for month t saw no information after month $t-1$. The 167-month sub-window from January 2011 onward overlaps the main test period reported in Section 5.1 and serves as a methodological cross-check; expanding-window OOS produces a Sharpe of 0.88 over that window, within the production result's bootstrap interval.

H.3 Factor Model Regressions

To test whether XGB's out-of-sample return is explained by known risk factors, monthly long-short returns are regressed on five standard factor models: CAPM, Fama–French three-factor, Carhart four-factor, Fama–French five-factor, and a six-factor specification that augments FF5 with the momentum factor (UMD). Each regression takes the form

$$r_t^{\mathrm{LS}} = \alpha + \boldsymbol{\beta}^\top \mathbf{f}_t + \varepsilon_t, \tag{31}$$

where $\mathbf{f}_t$ is the factor return vector for the specification in question, $\boldsymbol{\beta}$ measures the strategy's exposure to each factor, and α is the average return earned in excess of what those exposures would mechanically generate. The null hypothesis $\alpha = 0$ states that the factor model fully explains the strategy's returns; rejecting it means the strategy earns return that no linear combination of the factors can replicate. Newey–West standard errors (6 lags) are used throughout to correct for residual autocorrelation and heteroskedasticity in monthly returns. Table 21 reports $\hat{\alpha}$ and its t-statistic under each specification; Table 22 repeats the exercise for the fund-augmented variant. Alphas that survive all five factor models indicate return that cannot be attributed to market, size, value, profitability, investment, or momentum exposure.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	23.4	4.08	-0.14					
FF3	23.2	4.48	-0.14	+0.04	-0.17			
Carhart	24.3	4.75	-0.19	-0.01	-0.22	-0.20		
FF5	22.9	4.62	-0.13	+0.08	-0.23		+0.06	+0.12
FF6	24.1	4.81	-0.18	+0.01	-0.31	-0.22	+0.01	+0.21

Table 21: Factor model regressions for XGB (mom+π). α is annualised. t-statistics use Newey–West standard errors (6 lags).

For completeness, Table 22 reports the same regressions for the fund-augmented variant (XGB: mom+π+fund) discussed in Section 5.1.3. Alphas remain statistically significant under all five specifications ($t > 2.9$) but are materially smaller than the baseline's, consistent with the dilution documented in the main text.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	17.6	3.04	-0.16					
FF3	17.1	3.12	-0.14	-0.06	-0.13			
Carhart	18.2	3.22	-0.19	-0.12	-0.18	-0.20		
FF5	16.4	3.02	-0.11	+0.01	-0.30		+0.09	+0.35
FF6	17.7	3.17	-0.16	-0.06	-0.39	-0.23	+0.04	+0.44

Table 22: Factor model regressions for XGB (mom+π+fund). α is annualised. Newey–West t-statistics (6 lags).

Residual diagnostics

To verify that the 6-lag Newey–West truncation is empirically appropriate for our data, we apply the Ljung–Box portmanteau test to the residuals of each factor regression. Table 23 reports the Ljung–Box statistic at lag 6 and the p-values at lags 6 and 12 for every (strategy, model) pair; the test supports the 6-lag truncation for every entry in the table ($p > 0.05$ throughout). Figure 13 plots the residual autocorrelation function out to lag 24 for the six-factor regression of each strategy. Individual autocorrelations are small in magnitude; a small number of lags fall marginally outside the $\pm 1.96/\sqrt{T}$ bands (lag 6 for XGB at $\hat{\rho}_6 = 0.154$, for example) but none reaches Bonferroni-corrected significance across 24 lags, and the joint Ljung–Box test fails to reject. As a direct robustness check on the lag truncation, the t-statistic on the XGB six-factor α is stable across Newey–West truncation lags $L \in \{3, 6, 12, 24\}$, taking values $\{4.77, 4.81, 4.88, 4.74\}$. Whatever residual dependence exists at long lags is small enough that inference does not depend on the choice of truncation.

			LB(6)		LB(12)	
Strategy	Model	T	Q_6	p	Q_{12}	p
XGB	CAPM	167	8.23	0.221	15.59	0.211
XGB	FF3	167	9.45	0.150	17.61	0.128
XGB	Carhart	167	10.46	0.106	17.20	0.142
XGB	FF5	167	9.65	0.140	17.72	0.125
XGB	FF6	167	10.74	0.097	17.51	0.132
D&M	CAPM	167	2.94	0.817	5.57	0.936
D&M	FF3	167	2.82	0.831	5.31	0.947
D&M	Carhart	167	2.88	0.824	5.40	0.943
D&M	FF5	167	2.32	0.888	4.79	0.965
D&M	FF6	167	2.18	0.903	4.63	0.969
WML	CAPM	167	6.16	0.406	16.24	0.180
WML	FF3	167	7.46	0.281	18.18	0.110
WML	Carhart	167	6.84	0.336	18.13	0.112
WML	FF5	167	7.13	0.309	17.83	0.121
WML	FF6	167	6.87	0.333	17.82	0.121
LR	CAPM	167	2.67	0.849	18.99	0.089
LR	FF3	167	2.36	0.884	18.90	0.091
LR	Carhart	167	2.34	0.886	19.08	0.087
LR	FF5	167	2.71	0.844	17.71	0.125
LR	FF6	167	2.71	0.844	17.73	0.124

Table 23: Ljung–Box residual diagnostics for the factor regressions in Table 21. Q_6 and Q_{12} are the Ljung–Box statistics at lags 6 and 12, each reported with its p-value. Across every (strategy, model) pair we fail to reject the null of no residual autocorrelation at the 5% level. The Newey–West truncation of six lags used throughout the thesis is empirically supported.

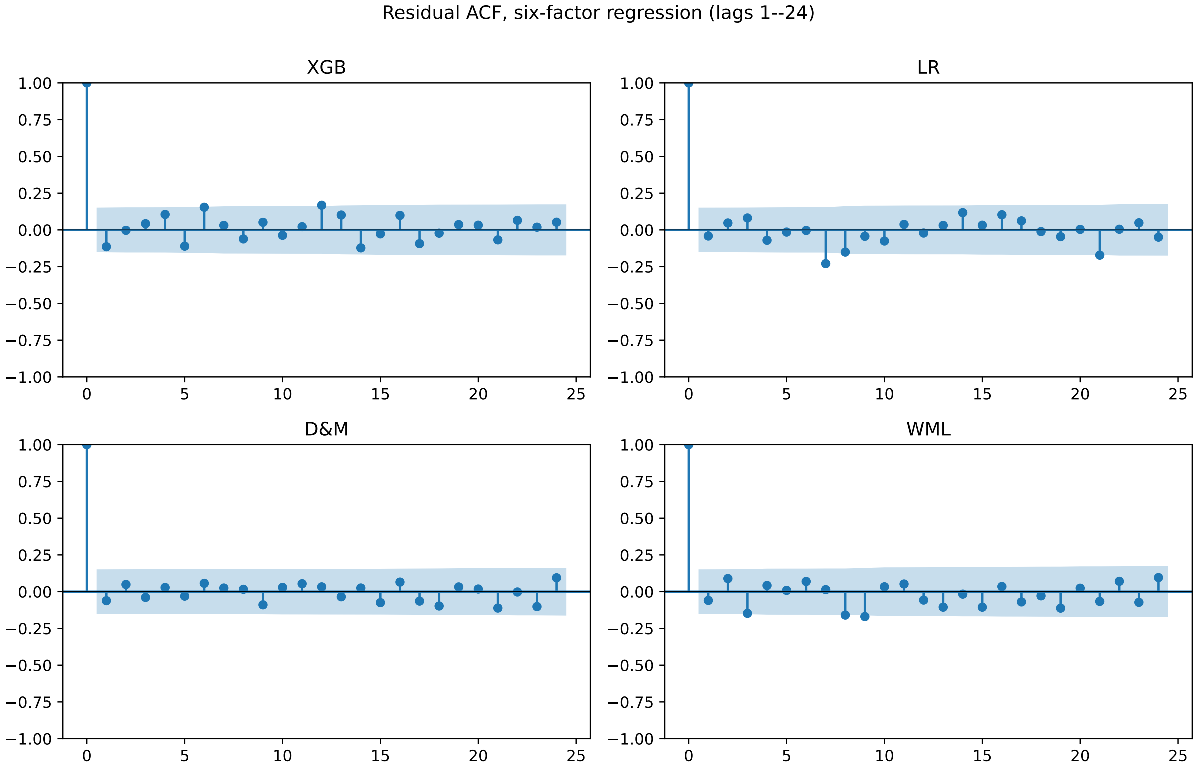

Figure 13: Residual autocorrelation function for the six-factor regression of each strategy. Shaded bands are the $\pm 1.96/\sqrt{T}$ confidence interval under the null of white-noise residuals. Individual autocorrelations are small in magnitude; the occasional lag sitting marginally outside the band does not reach significance after multiple-testing correction, consistent with the joint Ljung–Box result in Table 23.

H.4 Block Bootstrap Inference

To assess the sampling variability of XGB's Sharpe ratio and test its outperformance formally, a block bootstrap is applied to the monthly return series. Returns are resampled with replacement in 12-month blocks to preserve autocorrelation, producing 10,000 synthetic 167-month paths (fixed seed 42). The Sharpe ratio is recomputed on each resample to obtain a 95% confidence interval, and the same block draws are used for XGB and each benchmark to form a paired test of the Sharpe difference. The bottom panel repeats the exercise on XGB returns split by regime.

	Sharpe	95% CI	vs XGB (p)
XGB	1.11	$[0.67, 1.53]$	—
LR	−0.01	$[-0.41, 0.42]$	0.001***
DET (Formula)	0.11	$[-0.36, 0.59]$	0.008***
Fixed 12-mo mom	0.06	$[-0.43, 0.59]$	0.005***
Fixed 1-mo mom	0.26	$[-0.21, 0.73]$	0.021**
XGB regime-conditional Sharpe			
XGB Calm (n =108)	0.84	$[0.29, 1.32]$	—
XGB Panic (n =59)	1.54	$[0.85, 2.22]$	—
Panic − Calm	0.70	[-0.16, 1.59]	0.109

Table 24: Block bootstrap confidence intervals for Sharpe ratios and paired tests of XGB's outperformance. Uses 12-month blocks and 10,000 resamples with a fixed seed. The "vs XGB (p)" column reports two-sided p-values from paired tests where both strategies are resampled on the same dates. $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

XGB's Sharpe confidence interval is [0.67, 1.53], comfortably above zero, while every benchmark's interval straddles zero. The paired tests reject equality of Sharpe ratios between XGB and each benchmark at $p < 0.025$. Within XGB, the panic-period Sharpe exceeds the calm-period Sharpe in point estimate (1.53 vs 0.84), but with only 59 panic months the paired difference is not statistically significant ($p = 0.109$); the 95% interval $[-0.16, 1.59]$ straddles zero. The point-estimate gap is consistent with the regime-conditional selection mechanism described in Section 5.2, but its statistical confirmation would require a longer panic sample. These intervals describe sampling variability within the 2011–2024 sample; out-of-sample generalization is addressed separately by the 25-year expanding-window backtest (Section 5.3.2; methodology in Appendix H.2) and the sub-period analysis (Appendix G.1).

H.5 Leg-Level CAPM Betas by Regime

The D&M comparison in Section 5.3.3 characterizes XGB's contribution as stock selection rather than exposure scaling. A direct test of this framing is to compute leg-level CAPM betas separately by regime: if XGB reorganizes which stocks it holds across regimes (cross-sectional re-ranking), the long-leg and short-leg betas should shift between calm and panic in opposite directions; if instead XGB mechanically scales exposure, both legs' betas should move together.

Table 25: Leg-level CAPM betas by regime, 2011–2024 OOS test period. Panic months are those with $\pi_t^{\text{filter}} \geq 0.5$. Newey–West standard errors (3 lags) in parentheses.

Strategy	Leg	Full	Calm	Panic
Fixed 12-mo mom	Long	1.09 (0.06)	1.09 (0.09)	1.07 (0.08)
Fixed 12-mo mom	Short	1.74 (0.11)	1.80 (0.21)	1.69 (0.13)
XGB	Long	1.55 (0.08)	1.40 (0.11)	1.65 (0.10)
XGB	Short	1.10 (0.06)	1.22 (0.09)	0.98 (0.05)

The pattern in Table 25 is consistent with the cross-sectional re-ranking interpretation. XGB's long-leg beta rises from 1.40 in calm to 1.65 in panic (*tilt amplification*: in panic, the long leg holds names with higher market sensitivity), while the short-leg beta falls from 1.22 to 0.98. Fixed 12-month momentum, by contrast, holds both leg-level betas roughly stable across regimes (long: 1.09 calm, 1.07 panic; short: 1.80 calm, 1.69 panic), consistent with the unconditional rank-based construction. The difference is direct evidence that the regime signal acts on *which* stocks are held rather than on *how much*.

H.6 Per-cluster Feature Signature ($K = 4$ Partition)

Section 5.2.2 reports a $K = 4$ partition of the 167 test months and characterizes each cluster narratively. Table 26 provides the full per-cluster feature signature for completeness: cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each cluster's months; long-leg pick raw momentum at each horizon, averaged across each cluster's months; and long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 7 in the main text.

Feature	Cluster 1 *calm cont.*	Cluster 2 *mild cont.*	Cluster 3 *post-panic*	Cluster 4 *deep crisis*
Cross-section context				
Cross-section mom (overall mean)	13.8%	7.3%	13.5%	−9.3%
Cross-section mom (short tertile, $h = 1$–4)	6.1%	4.4%	3.0%	−7.1%
Cross-section mom (mid tertile, $h = 5$–8)	14.6%	7.6%	12.4%	−11.1%
Cross-section mom (long tertile, $h = 9$–12)	20.8%	9.7%	25.0%	−9.6%
Cross-section dispersion (short)	0.23	0.27	0.28	0.23
Cross-section dispersion (mid)	0.44	0.46	0.56	0.35
Cross-section dispersion (long)	0.60	0.59	0.85	0.45
Cross-section skewness (short)	4.6	7.2	6.4	3.7
Cross-section skewness (mid)	6.2	7.6	6.2	5.6
Cross-section skewness (long)	6.7	8.1	6.5	5.7
π^{filter} frequency, prior 6 months	0.27	0.37	0.30	0.51
π^{filter} frequency, prior 12 months	0.31	0.32	0.41	0.30
Strategy Sharpe, prior 12 months	1.22	1.04	1.15	1.08
Long-leg pick raw momentum by horizon				
Long-leg pick mom, $h = 1$	7.1%	3.1%	−3.4%	−15.3%
Long-leg pick mom, $h = 2$	13.5%	8.6%	−5.5%	−21.6%
Long-leg pick mom, $h = 3$	18.3%	8.8%	−5.3%	−26.3%
Long-leg pick mom, $h = 4$	23.1%	10.6%	−5.8%	−30.8%
Long-leg pick mom, $h = 5$	30.8%	13.8%	−5.9%	−35.0%
Long-leg pick mom, $h = 6$	40.2%	18.0%	−3.2%	−38.3%
Long-leg pick mom, $h = 7$	49.4%	20.7%	−1.3%	−42.1%
Long-leg pick mom, $h = 8$	59.2%	23.7%	0.0%	−44.9%
Long-leg pick mom, $h = 9$	61.3%	23.1%	1.1%	−46.5%
Long-leg pick mom, $h = 10$	60.7%	21.9%	1.3%	−47.2%
Long-leg pick mom, $h = 11$	61.1%	22.8%	1.2%	−47.3%
Long-leg pick mom, $h = 12$	57.9%	18.5%	−5.3%	−47.5%
Long-leg pick cross-sectional z-score by horizon				
Long-leg z-curve, $h = 1$	0.26	0.04	−0.28	−0.73
Long-leg z-curve, $h = 2$	0.38	0.14	−0.31	−0.74
Long-leg z-curve, $h = 3$	0.40	0.11	−0.34	−0.75
Long-leg z-curve, $h = 4$	0.44	0.10	−0.36	−0.80
Long-leg z-curve, $h = 5$	0.54	0.16	−0.35	−0.85
Long-leg z-curve, $h = 6$	0.65	0.23	−0.29	−0.83
Long-leg z-curve, $h = 7$	0.73	0.27	−0.26	−0.85
Long-leg z-curve, $h = 8$	0.80	0.32	−0.22	−0.87
Long-leg z-curve, $h = 9$	0.76	0.29	−0.23	−0.88
Long-leg z-curve, $h = 10$	0.72	0.24	−0.25	−0.87
Long-leg z-curve, $h = 11$	0.67	0.22	−0.26	−0.86
Long-leg z-curve, $h = 12$	0.54	0.12	−0.35	−0.87

Table 26: Per-cluster feature signature for the $K = 4$ partition, 2011–2024 test period. Features are grouped into three blocks: (1) cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each clusterś months; (2) long-leg pick raw momentum at each horizon, averaged across each clusterś months; (3) long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 7; this table is the full descriptive feature signature.

H.7 Cluster-Level Cross-Validation

Three diagnostics extend the regime-aggregate findings of Section 5.2 across the $K = 4$ partition.

Leg-betas across the $K = 4$ partition

XGB retains long-leg $\beta >$ short-leg β in every cluster (spreads +0.68, +0.40, +0.49, +0.43 for clusters 1–4), while unconditional 12-month momentum exhibits leg-beta inversion in every cluster (−0.24, −0.87, −0.75, −0.42; Table 27). Daniel and Moskowitz (2016)'s crash mechanism is latent across every cluster type; XGB escapes it everywhere through compositional re-ranking.

		XGB			Fixed 12-mo mom		
Cluster	n	β_L	β_S	Spread	β_L	β_S	Spread
1 (calm continuation)	53	1.65 (0.22)	0.97 (0.13)	0.68	1.32 (0.22)	1.57 (0.24)	−0.24
2 (transition)	62	1.60 (0.15)	1.20 (0.09)	0.40	1.04 (0.08)	1.91 (0.17)	−0.87
3 (post-panic recovery)	31	1.46 (0.16)	0.97 (0.11)	0.49	1.09 (0.17)	1.84 (0.22)	−0.75
4 (deep crisis)	21	1.49 (0.08)	1.06 (0.09)	0.43	1.07 (0.10)	1.49 (0.14)	−0.42

Table 27: Leg-level CAPM betas by K=4 cluster, 2011–2024 test period. XGB (regime-conditional XGBoost ensemble) versus unconditional 12-month momentum, side-by-side. Standard errors in parentheses. Spread = $\beta_L - \beta_S$.

π_t^{filter}'s SHAP share by cluster

Combined long-and-short SHAP shares are 0.50, 0.46, 0.54, 0.24 for clusters 1–4 (Table 28). These are consistent with the panic-aggregate 41% share reported in Section 5.2.1: the 41% averages across all panic months, and the cluster-level shares show that the collapse is concentrated in cluster 4 (deep panic, share 0.24), exactly where the main-text discussion attributes the decline to π_t^{filter} saturating near 1. The pattern is consistent with the routing interpretation: π_t^{filter} contributes most where the cross-section is ambiguous (cluster 3) and least where it is unambiguously crisis-shaped (cluster 4, where momentum splits alone suffice to produce the loser portfolio).

Cluster	n	Long π share	Short π share	Combined π share
1 (calm continuation)	53	34.0%	50.1%	49.9%
2 (transition)	62	26.9%	46.5%	46.0%
3 (post-panic recovery)	31	27.8%	53.7%	53.6%
4 (deep crisis)	21	11.4%	38.1%	24.4%

Table 28: π^{filter}'s SHAP share by K=4 cluster, 2011–2024 test period. Long and short shares are computed over long-leg and short-leg picks respectively; combined is over all stock-months in the cluster.

Short-leg structure (future work)

The $K = 4$ partition was defined on long-leg z-curves; applying the same labels to the short leg shows the short basket is structurally distinct from a negated long leg, with short-vs-negated-long correlations of $+0.94$, $+0.89$, $+0.19$, and -0.84 for clusters 1–4 (Table 29). The cluster-4 sign flip from positive to negative correlation means the model is not merely shorting the mirror image of its long-leg picks: it selects a qualitatively different basket whose cross-sectional fingerprint moves opposite to a mechanical "short the long-leg pattern" rule. The month-1 short-leg spike in cluster 4 ($+0.57$) is consistent with Nagel (2012)'s finding that short-horizon reversal returns spike during high-volatility periods: month 1 carries 10.6% of panic short-leg momentum SHAP versus 5.3% in calm. A full short-leg analysis is left to future work.

Cluster	n	z_1	$\bar{z}$	Basket	$\rho(-z^L)$
1 (calm continuation)	53	−0.062	−0.365	391	0.943
2 (transition)	62	0.101	−0.174	334	0.888
3 (post-panic recovery)	31	0.209	0.241	382	0.190
4 (deep crisis)	21	0.568	0.170	353	−0.835

Table 29: Short-leg z-curve summary by K=4 cluster, 2011–2024 test period. z_1 is the cluster-mean cross-sectional z-score of short-leg picks at the 1-month horizon; $\bar{z}$ averages across all 12 horizons. Basket size is mean stocks per month. $\rho(-z^L)$ is the correlation between the short-leg z-curve centroid and the negated long-leg z-curve centroid; values below 0.95 indicate structural asymmetry.

H.8 Emission Distribution Robustness

To verify that regime identification is robust to the choice of emission distribution, the HMM is re-estimated with multivariate Student-t emissions across a grid of degrees of freedom $\nu \in \{3, 5, 7, 10, 15, 20, 30, 50, 100\}$.

The Student-t model is estimated via data augmentation (Geweke, 1993): each observation receives a latent weight $w_t \sim \text{Gamma}(\nu/2, \nu/2)$, so that $\mathbf{z}_t \mid s_t{=}k, w_t \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k/w_t)$. Marginalising over w_t yields a multivariate t with ν degrees of freedom. Conditional on w_t the emission is Gaussian, so the Normal-Inverse-Wishart conjugate updates carry over with weighted sufficient statistics, preserving the efficiency of the Gibbs sampler.

ν	LL (train)	LL (test)	BIC	Agreement	Corr(π)
3	−882.7	−701.1	1929.9	91.4%	0.872
5	−829.0	−669.8	1822.5	96.8%	0.960
7	−838.3	−698.2	1841.2	98.0%	0.984
10	−815.8	−644.3	1796.1	96.8%	0.960
15	−800.1	−651.0	1764.8	98.3%	0.997
20	−802.9	−621.7	1770.4	99.3%	0.997
30	−802.6	−640.6	1769.8	98.5%	0.993
50	−796.6	−658.1	1757.7	98.8%	0.996
100	−792.3	−643.5	1749.0	98.8%	0.987
Normal	−787.5	−642.5	**1739.5**	—	—

Table 30: Student-t vs. Normal emission HMM comparison. LL = log-likelihood; Agreement = fraction of months with identical binary regime classification as the Normal HMM; Corr(π) = correlation of filtered panic probabilities.

The Normal model achieves the lowest BIC (2257.3 vs. 2257.6 for the best Student-t at $\nu = 100$), indicating no meaningful improvement from heavier tails. Regime classifications agree with the Normal HMM on at least 97.8% of months across all ν values, and the correlation of filtered panic probabilities exceeds 0.98 everywhere.

Appendix I: Label Sign Correction

Because the four HMM features respond to stress in different directions, the expected direction of each feature during stress is determined empirically. For each feature j, the 95th percentile is computed separately on training months falling within known crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) and on all remaining training months:

$$\mathrm{sign}_j = \begin{cases} +1 & \text{if } q_{0.95}^{\text{crisis}}(z_j) \geq q_{0.95}^{\text{normal}}(z_j), \\ -1 & \text{otherwise.} \end{cases} \tag{32}$$

Features that spike upward during crises receive $\mathrm{sign}_j = +1$; features that decline receive $\mathrm{sign}_j = -1$. The panic state is then identified as the regime $k \in \{0, 1\}$ whose posterior mean vector $\hat{\boldsymbol{\mu}}_k$ best aligns with the crisis direction:

$$k^* = \arg\max_{k \in \{0,1\}} \sum_{j=1}^{D} \mathrm{sign}_j \cdot \hat{\mu}_{k,j}. \tag{33}$$

The state labeled k^* is designated as "panic" and the other as "calm."

Appendix J: Convergence Diagnostics

Convergence of the Gibbs sampler is assessed in two complementary ways: effective sample sizes (ESS) within each chain, and the Gelman–Rubin $\hat{R}$ statistic across independent chains. The two are complementary because ESS measures efficiency within a single chain but cannot detect whether different chains have converged to different posterior modes, while Gelman–Rubin compares chains but says nothing about within-chain autocorrelation. Trace plots provide a third, visual check.

J.1 Effective Sample Size (ESS)

Because successive MCMC draws are autocorrelated, n draws from the chain contain less information than n independent samples. The ESS quantifies the number of truly independent draws the chain is equivalent to. For a post-burn-in chain of length n,

$$ESS = \frac{n}{1 + 2\sum_{\ell=1}^{L^*} \hat{\rho}(\ell)}, \tag{34}$$

where $\hat{\rho}(\ell)$ is the estimated autocorrelation at lag ℓ,[25] and L^* is the first lag at which the autocorrelation becomes negative. High ESS indicates that the chain provides many effectively independent posterior draws; an ESS of at least 100 per parameter is a common minimum for reliable posterior inference (Gelman et al., 2013).

[25] $\hat{\rho}(\ell) = \frac{1}{(n-\ell)\,\hat{\sigma}^2} \sum_{i=1}^{n-\ell} (\theta_i - \bar{\theta})(\theta_{i+\ell} - \bar{\theta})$, where θ_i is the i-th post-burn-in draw, $\bar{\theta}$ its sample mean, and $\hat{\sigma}^2$ its sample variance.

J.2 Trace plots

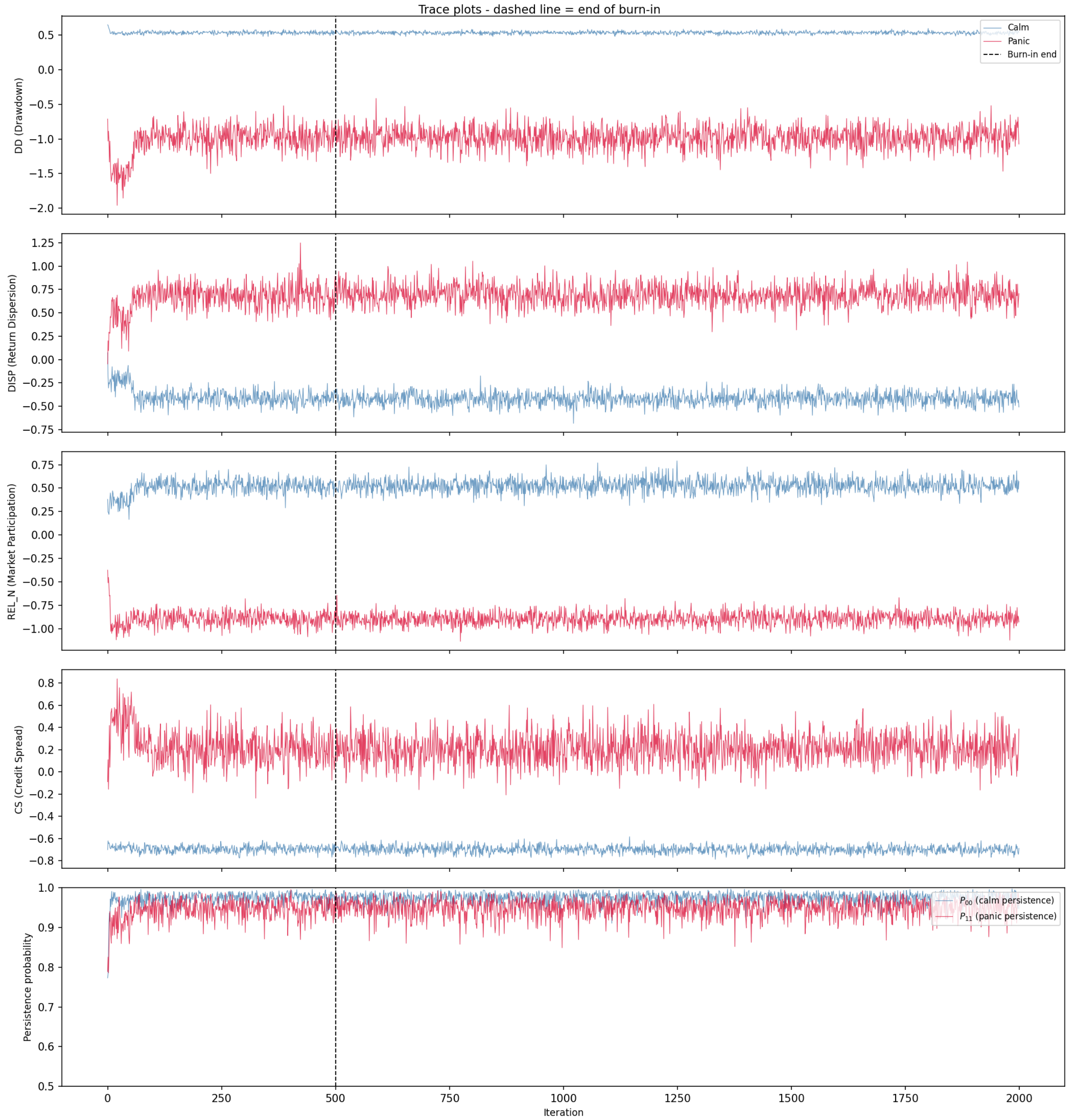

Figure 14: Trace plots of the Gibbs sampler. Each panel shows sampled regime means (calm in blue, panic in red) for one HMM feature. The dashed line marks the end of burn-in.

J.3 Gelman–Rubin Diagnostic

The Gelman–Rubin $\hat{R}$ statistic (Gelman and Rubin, 1992; Gelman et al., 2013) compares within-chain and between-chain variance across M independent chains, each of length n.

Let s_m^2 denote the variance of chain m and $\bar{\theta}_m$ its mean. The within-chain variance is

$$W = \frac{1}{M} \sum_{m=1}^{M} s_m^2, \tag{35}$$

and the between-chain variance is

$$B = \frac{n}{M-1} \sum_{m=1}^{M} (\bar{\theta}_m - \bar{\theta})^2, \tag{36}$$

where $\bar{\theta} = \frac{1}{M} \sum_m \bar{\theta}_m$. If all chains have converged to the same posterior, B should be small relative to W. The diagnostic combines these:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n} W + \frac{B}{n}. \tag{37}$$

Values of $\hat{R}$ near 1.0 indicate convergence; the conventional threshold is $\hat{R} < 1.1$, with a stricter modern bound at $\hat{R} < 1.01$.

Parameter	$\hat{R}$	Status
μ_{panic}(DD)	1.000	OK
μ_{panic}(DISP)	1.000	OK
μ_{panic}(REL_N)	1.000	OK
μ_{panic}(CS)	1.000	OK
μ_{calm}(DD)	1.000	OK
μ_{calm}(DISP)	1.001	OK
μ_{calm}(REL_N)	1.000	OK
μ_{calm}(CS)	1.000	OK
P_{00} (calm persistence)	1.000	OK
P_{01} (calm $\rightarrow$ panic)	1.000	OK
P_{10} (panic $\rightarrow$ calm)	1.000	OK
P_{11} (panic persistence)	1.000	OK

Table 31: Gelman–Rubin $\hat{R}$ statistics for all HMM parameters, computed across 5 independent chains (1,500 post-burn-in draws each). All values are below 1.01, well within the convergence threshold of 1.1.

J.4 Seed Convergence of the Regime Signal

Gelman–Rubin and the trace plots assess whether a single Gibbs chain has converged to its stationary posterior, but they do not quantify how many chains are needed to stabilize the *downstream* Sharpe ratio. The regime signal π_t^{filter} enters the pipeline as the average of N_s seed-specific forward filters (Section 3.1.5), and because XGBoost is a nonlinear function of π_t^{filter}, MCMC noise in any individual chain does not cancel linearly.

To measure this dependence, we subsample $k \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 75, 100\}$

chains without replacement from the seed pool, average the forward filter over those k chains, retrain the XGBoost ensemble on the resulting signal, and record the out-of-sample Sharpe. For each $k < 100$ we repeat on 30 random subsets; $k = 100$ exhausts the pool.

k	n_{sub}	Mean	Median	Std	P25	P75	Min	Max
1	30	1.040	1.031	0.078	0.985	1.090	0.915	1.205
2	30	1.062	1.081	0.069	1.019	1.103	0.894	1.217
3	30	1.049	1.044	0.043	1.014	1.075	0.973	1.143
5	30	1.042	1.042	0.047	1.002	1.083	0.967	1.154
10	30	1.053	1.050	0.036	1.038	1.066	0.965	1.145
15	30	1.075	1.079	0.029	1.055	1.097	1.008	1.139
20	30	1.078	1.080	0.022	1.065	1.090	1.025	1.122
30	30	1.089	1.082	0.025	1.069	1.109	1.052	1.137
50	30	1.093	1.092	0.015	1.082	1.101	1.067	1.124
75	30	1.098	1.102	0.012	1.091	1.105	1.067	1.118
100	1	1.118	1.118	0.000	1.118	1.118	1.118	1.118

Table 32: HMM Gibbs-chain seed convergence of the downstream L/S Sharpe ratio. For each subset size k, we draw n_{sub} random subsets of k HMM seeds from a 100-seed pool, average their forward-filter outputs into π_t^{filter}, retrain the XGBoost ensemble on the averaged signal, and report the out-of-sample L/S Sharpe. The production configuration uses $N_s = 200$ HMM seeds; the 100-seed pool here is the subset over which the experiment is run. Mean Sharpe stabilizes around $k \approx$ 30–50; standard deviation across subsets shrinks as $1/\sqrt{k}$.

Two patterns emerge. First, the mean Sharpe rises from 1.04 at $k = 1$ to 1.12 at $k = 100$, so a single-chain point estimate understates the strategy's performance by roughly 0.08 in Sharpe units. The gap closes approximately monotonically (with minor non-monotonic dips at $k = 3$ and $k = 5$ before the trend resumes), and most of the lift is realized by $k = 30$. Second, the interquartile range across random subsets collapses from 0.10 at $k = 1$ to 0.02 at $k = 50$: once $k \geq 50$, any two random subsets of the seed pool produce nearly the same XGBoost portfolio. The production configuration ($N_s = 200$) sits well inside this plateau, which is why the 1.11 Sharpe reported in Section 5 is insensitive to which specific seeds were drawn. The practical implication is methodological rather than cosmetic: a single-chain HMM would not reliably support the main conclusions, so the seed ensemble is a requirement of the design, not a late-stage performance tuning.

Appendix K: International HMM Feature Selection

For the international robustness check (Section 5.3.4), the HMM feature set is selected per region using the same 4-pass procedure as the US production specification (Section 4.1): a quality screen on MCMC convergence and crisis-window alignment, a portfolio-Sharpe ranking on the surviving combinations, a definitive validation at higher seed counts, and an additional stability check across random seed partitions. The international candidate pool comprises the five locally-available stress features (DD, VOL, DISP, REL_N, BANK_REL); BANK_REL is the value-weighted bank-sector excess return over the broad regional market, sign-flipped so high values indicate stress, and substitutes for the US-specific Moody's BAA–AAA spread. With DD required, the procedure evaluates fifteen combinations of size one to four. Both UK and Japan select *DD+VOL+REL_N* as the regional feature set. The transplanted-from-US template (DD+DISP+REL_N+BANK_REL) fails the Pass-1 quality screen for both regions (minimum effective sample size of 14 and 4 respectively), motivating per-region selection. Pass-by-pass CSV results and chosen-features JSON manifests are saved under `results/thesis/` with the prefix `intl_<region>_hmm`.

Appendix L: Fundamental Feature Definitions

Ten firm-level characteristics are computed from Compustat quarterly data (`fundq`), linked to CRSP via the CCM bridge table. All fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. Cash-flow items (`oancfy`, `capxy`) are reported year-to-date and are differenced within each fiscal year to recover the per-quarter contribution before the trailing twelve-month sum is taken.

- **Book-to-market (bm):** Book equity (`ceqq`) divided by market equity.
- **Return on equity (roe):** Income before extraordinary items (`ibq`) divided by book equity.
- **Gross profitability:** Revenue minus cost of goods sold, scaled by total assets: $(\texttt{revtq} - \texttt{cogsq})/\texttt{atq}$.
- **Asset growth:** Year-over-year growth in total assets: $\texttt{atq}_t/\texttt{atq}_{t-4} - 1$.
- **Leverage:** Total debt divided by total assets: $(\texttt{dlttq} + \texttt{dlcq})/\texttt{atq}$.
- **Earnings growth:** Year-over-year growth in earnings, transformed as $\text{sign}(g) \cdot \log(1 + |g|)$.
- **Log market equity:** Natural logarithm of market capitalisation (price $\times$ shares outstanding), lagged one month.
- **Operating cash flow over assets (cfo_a):** Trailing twelve-month operating cash flow (`oancfy`, converted to quarterly and summed) scaled by total assets.
- **Free cash flow over assets (fcf_a):** Trailing twelve-month operating cash flow minus trailing twelve-month capital expenditure (`capxy`), scaled by total assets.

- **Accruals:** The accrual component of earnings, $(\texttt{ibq}_{\mathrm{TTM}} - \texttt{oancfy}_{\mathrm{TTM}})/\texttt{atq}$, which captures the non-cash portion of reported earnings.

Fundamentals are winsorized at the 1st and 99th percentiles (5th and 95th for asset growth) using training-sample bounds. The cash-flow features (cfo_a, fcf_a, accruals) are available for roughly 83% of stock-months; the remaining seven characteristics exceed 82% coverage individually, and XGBoost handles any residual missingness natively via its default-branch logic.